

Eigenvector distribution of random matrices under critical finite-rank deformations

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Abstract. We investigate the eigenvector distribution at the soft edge for Gaussian random matrices with finite-rank deformations, in the critical regime of BBP transition. For finite-rank deformations of the GOE and GUE with critical spikes, we find that the squared overlap between a leading eigenvector and a spike, rescaled by $N^{1/3}$, converges weakly to the negative reciprocal of the derivative of an Airy-Green function evaluated at the corresponding soft-edge root. For the rank-one critically spiked Gaussian β -ensemble, $\beta > 0$, we obtain an analogous result involving an Airy-Green function. In both cases, the Airy-Green functions are generalizations of the one introduced by Bykhovskaya–Gorin–Sodin [19]. The proofs are both based on an eigenvector–eigenvalue identity and a resolvent-differentiation mechanism.

Date: August 10, 2026

Keywords: eigenvector overlap, finite-rank deformation, Airy-Green function, stochastic Airy operator

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1. INTRODUCTION

Large random matrices with finite-rank deformation have been a widely used and well-studied model from both theoretical and applied perspectives in random matrix theory. A well-known result is the Baik-Ben Arous-Péché (BBP) phase transition [4, 38], which depicts a phase transition of the distribution of the leading eigenvalues of spiked complex Wishart matrices and spiked Gaussian unitary ensemble (GUE). In particular, the distribution in the critical regime, as an interpolation between the Tracy-Widom distribution and the Gaussian distribution, was discovered therein. The critical distribution in the real case, and a further extension to more general β -ensembles, was later given in [15, 16] as the smallest eigenvalues of the stochastic Airy operator with a certain boundary condition. Very recently, the critical eigenvalue was alternatively represented as a root of the so-called Airy-Green function in [19], for the real case.

Apart from the mentioned results on the critical eigenvalue distributions for the Wishart and Gaussian ensembles, there have been a vast number of results on the subcritical and supercritical regimes, and also extensions to more generally distributed random matrices with low-rank deformations in all regimes. Without being comprehensive, we refer to [3, 6, 7, 11, 22, 27, 31, 32] and the references therein.

Along with the BBP transition for the leading eigenvalues, there is a corresponding transition for the leading eigenvectors. In particular, in the supercritical regime, the top eigenvectors have a non-negligible overlap with the spike eigenvectors, while in the subcritical regime, they are nearly orthogonal; see [12] for instance. Results on finer scales including precise fluctuation results have also been obtained in the subcritical and supercritical regimes under rather general distributional assumptions of the matrix entries; see [14, 9, 8, 21] for instance. We also refer to [13, 20, 37, 35, 24], etc., for related studies.

In contrast, as an analogue of the main BBP result for the critical eigenvalue distribution in [4, 38], the eigenvector distribution in the critical regime has been much less studied. On the fluctuation level, the only available result was obtained in [10], for the spiked GUE model. In [10], using an eigenvector-eigenvalue identity, together with the determinantal structure of the GUE minor process with an external source, the eigenvector distribution can be represented by an extended Airy process. However, such a representation cannot be extended to spiked GOE due to the lack of an explicit distribution of the GOE minor process with an external source, not to mention the spiked Gaussian β -ensemble for general $\beta > 0$. In this work, we aim to introduce a unified representation for the leading eigenvector of the Gaussian orthogonal/unitary ensembles with finite-rank deformation by using the Airy-Green function introduced recently in [19], and extend the result to the Gaussian β -ensemble for general β , in the rank one case.

Below, we first introduce some preliminaries and notation in Section 1.1, and then state our main results in Section 1.2. We also record in Proposition 1.3 the almost-sure simplicity of the higher-rank Airy_1 and Airy_2 point processes, as an auxiliary consequence of the arguments in Sections 2.3 and 2.4.

1.1. Preliminaries and notations. In this subsection we introduce our matrix models and collect some necessary inputs used throughout the paper. We first recall the classical Gaussian ensembles

and the Dumitriu–Edelman tridiagonal Gaussian β -ensemble, and their finite-rank deformations. Further, we state some results regarding the stochastic Airy operator and the Airy-Green function.

Most of the results stated in this subsection are known in literature. The proofs of the new properties of the Airy-Green functions stated in this subsection, in particular Propositions 1.5, 1.6, and Lemma 1.7, are deferred to Section 4.

1.1.1. *Gaussian orthogonal ensemble, Gaussian unitary ensemble and Gaussian β -ensemble.* Let $H_N = (h_{ij})_{1 \leq i, j \leq N}$ be an $N \times N$ random matrix. We say that H_N is a *Gaussian Orthogonal Ensemble* (GOE) matrix if H_N is real symmetric,

$$h_{kk} \sim \mathcal{N}\left(0, \frac{2}{N}\right), \quad 1 \leq k \leq N, \quad h_{ij} \sim \mathcal{N}\left(0, \frac{1}{N}\right), \quad 1 \leq i < j \leq N, \quad (1.1)$$

and all diagonal and upper-triangular entries are independent. We say that H_N is a *Gaussian Unitary Ensemble* (GUE) matrix if H_N is complex Hermitian,

$$h_{kk} \sim \mathcal{N}\left(0, \frac{1}{N}\right), \quad 1 \leq k \leq N, \quad h_{ij} \sim \mathcal{N}\left(0, \frac{1}{2N}\right) + i\mathcal{N}\left(0, \frac{1}{2N}\right), \quad 1 \leq i < j \leq N, \quad (1.2)$$

and all diagonal and upper-triangular entries are independent. In either case, we write

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_N \quad (1.3)$$

for the eigenvalues of H_N in descending order.

For the above GOE and GUE matrices, the joint probability density function (j.p.d.f.) of $\{\mu_j\}_{j=1}^N$ is

$$\frac{1}{Z_{N,\beta}} \prod_{1 \leq i < j \leq N} |\mu_i - \mu_j|^\beta \exp\left(-\frac{\beta N}{4} \sum_{i=1}^N \mu_i^2\right), \quad (1.4)$$

with $\beta = 1$ in the GOE case and $\beta = 2$ in the GUE case.

For general $\beta > 0$, the N -dimensional *Gaussian β -ensemble* (G β E) is realized by the Dumitriu–Edelman tridiagonal random matrix [26]

$$H_N^{(\beta)} = \frac{1}{\sqrt{\beta N}} \begin{pmatrix} a_1^{(\beta)} & b_1^{(\beta)} & & & \\ b_1^{(\beta)} & a_2^{(\beta)} & b_2^{(\beta)} & & \\ & b_2^{(\beta)} & a_3^{(\beta)} & \ddots & \\ & & \ddots & \ddots & b_{N-1}^{(\beta)} \\ & & & b_{N-1}^{(\beta)} & a_N^{(\beta)} \end{pmatrix}, \quad (1.5)$$

where the entries are independent and satisfy

$$a_j^{(\beta)} \sim \mathcal{N}(0, 2), \quad 1 \leq j \leq N, \quad b_k^{(\beta)} \sim \chi_{\beta(N-k)}, \quad 1 \leq k \leq N-1.$$

The j.p.d.f. of the eigenvalues of $H_N^{(\beta)}$ for general $\beta > 0$ is given by (1.4); see [26]. We denote the eigenvalues of $H_N^{(\beta)}$ in descending order by

$$\mu_1^{(\beta)} \geq \mu_2^{(\beta)} \geq \cdots \geq \mu_N^{(\beta)}. \quad (1.6)$$

For $\beta = 1, 2$, the distribution of $\{\mu_j^{(\beta)}\}_{j=1}^N$ coincides with the GOE/GUE eigenvalue law.

The empirical spectral measure of a GOE/GUE matrix H_N , and more generally that of $H_N^{(\beta)}$, converges almost surely to the semicircle law

$$d\mu_{\text{SC}}(x) = \frac{\sqrt{4-x^2}}{2\pi} dx, \quad x \in [-2, 2].$$

The Stieltjes transform of semicircle law is

$$G_{\text{SC}}(z) = \int_{\mathbb{R}} \frac{1}{z-x} d\mu_{\text{SC}}(x) = \frac{z - \sqrt{z^2 - 4}}{2}, \quad z \in \mathbb{C} \setminus [-2, 2], \quad (1.7)$$

where the branch is chosen so that $\sqrt{z^2 - 4} \sim z$ as $|z| \rightarrow \infty$. Moreover, the top eigenvalue converges almost surely to 2; see [17] for instance.

1.1.2. Spiked deformations. Throughout the paper, we equip $\mathbb{R}^N$ with the standard Euclidean metric and $\mathbb{C}^N$ with the standard Hermitian metric. Let

$$e_i = (0, \dots, \underset{i\text{-th}}{1}, 0, \dots, 0)^\top, \quad 1 \leq i \leq N,$$

be the canonical basis vectors of $\mathbb{R}^N$ or $\mathbb{C}^N$, according to the ensemble under consideration, and $\langle \cdot, \cdot \rangle$ denotes the standard Euclidean or Hermitian inner product.

We consider additive finite-rank deformations of H_N or $H_N^{(\beta)}$ in the form of

$$Y = X + P, \quad \text{where} \quad P = \sum_{i=1}^r d_i \mathbf{w}_i \mathbf{w}_i^*. \quad (1.8)$$

When X is either a GOE or GUE matrix H_N , r can be any fixed positive integer and the vectors $\mathbf{w}_1, \dots, \mathbf{w}_r$ are deterministic orthonormal vectors in $\mathbb{R}^N$ in the real case and in $\mathbb{C}^N$ in the complex case. When X is $H_N^{(\beta)}$ and we take $r = 1$ and $\mathbf{w}_1 = e_1$. Denote the eigenvalues of Y by

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N,$$

and let $\mathbf{x}_k$ be a unit eigenvector associated with λ_k . Whenever the spectrum of Y is simple, $\mathbf{x}_k$ is determined by Y and λ_k up to a sign in the real case and up to a phase in the complex case. This simplicity holds almost surely in the cases considered below. Hence the overlaps

$$|\langle \mathbf{x}_k, \mathbf{w}_j \rangle|^2, \quad j = 1, \dots, r, \quad (1.9)$$

are unambiguously defined. For simplicity, we will call the quantities in (1.9) the eigenvector overlaps in the sequel.

Remark 1.1. Since GOE/GUE is orthogonally/unitarily invariant, the joint law of the eigenvector overlaps is unchanged if we replace $\mathbf{w}_j$'s by e_j 's.

Remark 1.2. When $\beta = 1, 2$ the eigenvector overlap law for $H_N^{(\beta)} + d_1 e_1 e_1^*$ is the same as that for $H_N + d_1 e_1 e_1^*$, where H_N is the GOE matrix for $\beta = 1$ and the GUE matrix for $\beta = 2$. Indeed, the Dumitriu–Edelman/Lanczos tridiagonalization is implemented by an orthogonal conjugation in the GOE case and by a unitary conjugation in the GUE case. In both cases the conjugating matrix can be chosen to fix e_1 , and therefore it leaves $e_1 e_1^*$ unchanged. The eigenvectors are conjugated, but

$$|\langle \mathbf{x}_k, e_1 \rangle|$$

is preserved. Hence the rank-one overlap law of the invariant GOE/GUE model coincides with that of the tridiagonal model $H_N^{(\beta)} + d_1 e_1 e_1^*$ for $\beta = 1, 2$.

1.1.3. *Scalar and multivariate stochastic Airy operators.* Let $b = (b_x)_{x \geq 0}$ be a standard real Brownian motion. For $\beta > 0$, the scalar stochastic Airy operator is formally defined as

$$\mathcal{H}_\beta = -\frac{d^2}{dx^2} + x + \frac{2}{\sqrt{\beta}} b'_x, \quad (1.10)$$

on $(0, \infty)$, where b'_x is real white noise, understood as the distributional derivative of b_x . For $\theta \in \mathbb{R} \cup \{\infty\}$, we impose the Robin boundary condition

$$f'(0) = \theta f(0),$$

with the convention that $\theta = \infty$ means the Dirichlet boundary condition $f(0) = 0$.

The construction of $\mathcal{H}_\beta$ with Dirichlet boundary condition was established in [39]. The Robin boundary condition

$$f'(0) = \theta f(0)$$

is treated in [15, Section 2]. The existence and discreteness of the spectrum follow from [29, Theorem 2.2 and Corollary 2.3], while its simplicity follows from [15, Lemma 2.7].

Proposition 1.1. *Fix $\beta > 0$. For every $\theta \in \mathbb{R} \cup \{\infty\}$, the operator $\mathcal{H}_\beta$ with boundary condition $f'(0) = \theta f(0)$, interpreted as $f(0) = 0$ when $\theta = \infty$, has, almost surely, a purely discrete, simple spectrum bounded from below. We enumerate its eigenvalues in strictly increasing order as*

$$\Lambda_{\beta,1}^{(\theta)} < \Lambda_{\beta,2}^{(\theta)} < \cdots, \quad \Lambda_{\beta,k}^{(\theta)} \longrightarrow +\infty.$$

The following convergence statements are essentially from [15, Theorem 1.5] and [39, Theorem 1.1]. In particular, the unperturbed Dirichlet case, i.e., $\theta = \infty$, is equivalent to the soft-edge limit in [39, Theorem 1.1].

Proposition 1.2. *Fix $\beta > 0$, $\theta \in \mathbb{R}$, and $k \geq 1$. Let $\mu_1^{(\beta)} \geq \cdots \geq \mu_N^{(\beta)}$ be the eigenvalues of $H_N^{(\beta)}$, and let $\lambda_1 \geq \cdots \geq \lambda_N$ be the eigenvalues of*

$$H_N^{(\beta)} + d e_1 e_1^\top, \quad d = 1 - \theta N^{-1/3}.$$

Then

$$(N^{2/3}(2 - \mu_1^{(\beta)}), \dots, N^{2/3}(2 - \mu_k^{(\beta)})) \implies (\Lambda_{\beta,1}^{(\infty)}, \dots, \Lambda_{\beta,k}^{(\infty)}),$$

and

$$(N^{2/3}(2 - \lambda_1), \dots, N^{2/3}(2 - \lambda_k)) \implies (\Lambda_{\beta,1}^{(\theta)}, \dots, \Lambda_{\beta,k}^{(\theta)}).$$

In the case of spiked GOE/GUE with rank $m > 1$ deformation, one needs the multivariate stochastic Airy operator introduced by Bloemendal and Virág [16, Section 1]. This operator describes the soft-edge limit of the finite-rank invariant model after keeping track of the whole m -dimensional spiked subspace. At the discrete level, Bloemendal and Virág introduce a canonical $(2m+1)$ -diagonal band form. After grouping the coordinates into blocks of size m , this form may equivalently be viewed as a block-tridiagonal matrix.

Let $\mathbb{F} = \mathbb{R}$ for $\beta = 1$ and $\mathbb{F} = \mathbb{C}$ for $\beta = 2$. A standard matrix Brownian motion $B = (B_x)_{x \geq 0}$ in the self-adjoint $m \times m$ matrices over $\mathbb{F}$ is normalized so that, for $0 \leq y \leq x$, $B_x - B_y$ has the law of $\sqrt{m(x-y)}$ times an $m \times m$ GOE matrix when $\beta = 1$, and $\sqrt{m(x-y)}$ times an $m \times m$ GUE matrix when $\beta = 2$, where the GOE/GUE matrices have the normalization specified above.

Equivalently, for $\beta = 1$, the diagonal entries are independent Brownian motions with variance parameter 2, and the upper-triangular off-diagonal entries are independent standard real Brownian motions. For $\beta = 2$, the diagonal entries are independent standard real Brownian motions,

and the upper-triangular off-diagonal entries are independent standard complex Brownian motions satisfying

$$\mathbb{E} |B_{ij}(x) - B_{ij}(y)|^2 = x - y.$$

All these processes are mutually independent, subject to self-adjoint symmetry.

$$\mathcal{H}_{\beta,m} = -\frac{d^2}{dx^2} + mx I_m + \sqrt{2} B'_x,$$

acting on $L^2(\mathbb{R}_+; \mathbb{F}^m)$. We impose the self-adjoint boundary condition

$$f'(0) = W f(0),$$

where W is real symmetric for $\beta = 1$ and Hermitian for $\beta = 2$. Since the law of B_x is invariant under fixed orthogonal/unitary conjugation, the law of the spectrum depends on W only through its eigenvalues. Hence, for distributional purposes, we may assume that

$$W = \text{diag}(\theta_1, \dots, \theta_m).$$

Writing

$$\Theta = (\theta_1, \dots, \theta_m) \in \mathbb{R}^m,$$

the boundary condition becomes

$$f'_i(0) = \theta_i f_i(0), \quad i = 1, \dots, m. \quad (1.11)$$

This is precisely the Robin boundary convention used in [16, Section 1, Equations (1.1)–(1.2)]. We record the spectral properties of $\mathcal{H}_{\beta,m}$ that will be used below.

Proposition 1.3. *Let $m \in \mathbb{N}$ and let $\Theta = (\theta_1, \dots, \theta_m) \in \mathbb{R}^m$ be fixed. For $\beta = 1, 2$, the operator $\mathcal{H}_{\beta,m}$ with boundary condition (1.11) has, almost surely, a purely discrete, simple spectrum bounded from below. We write its eigenvalues in strictly increasing order as*

$$\Lambda_{\beta,1}^\Theta < \Lambda_{\beta,2}^\Theta < \dots, \quad \Lambda_{\beta,k}^\Theta \longrightarrow +\infty.$$

Remark 1.3. The lower boundedness of the spectrum in Proposition 1.3 follows from the quadratic-form estimate [16, Equation (3.16)], while its existence and discreteness follow from [16, Proposition 3.7 and Remark 3.8]. In the scalar case $m = 1$, simplicity follows from [15, Lemma 2.7]. For $m \geq 2$, simplicity is established as part of our proofs in Sections 2.3 and 2.4 for $\beta = 1$ and $\beta = 2$, respectively. We note that the simplicity conclusion can also be deduced from the very recent multilevel construction of Landon and Stojimirovic [33, Equations (1.7)–(1.8), Proposition 1.1, and Theorem 1.2], in which the successive limiting point processes are strictly interlacing.

When $m = 1$ and $\Theta = (\theta)$, the operator $\mathcal{H}_{\beta,1}$ coincides with $\mathcal{H}_\beta$ defined in (1.10) with Robin boundary parameter θ , for $\beta = 1, 2$. The next proposition is a rescaled special case of [16, Theorem 1.2] for $\beta = 1, 2$.

Proposition 1.4. *Let H_N be a GOE matrix when $\beta = 1$ and a GUE matrix when $\beta = 2$. Let*

$$P = \sum_{i=1}^m d_i e_i e_i^* = \text{diag}(d_1, \dots, d_m, 0, \dots, 0), \quad d_i = 1 - \theta_i N^{-1/3}, \quad i = 1, \dots, m,$$

with $\theta_i \in \mathbb{R}$. Write

$$\lambda_1 \geq \dots \geq \lambda_N$$

for the eigenvalues of $H_N + P$. Then, for each fixed $k \geq 1$,

$$(N^{2/3}(2 - \lambda_1), \dots, N^{2/3}(2 - \lambda_k)) \implies (\Lambda_{\beta,1}^{(\theta_1, \dots, \theta_m)}, \dots, \Lambda_{\beta,k}^{(\theta_1, \dots, \theta_m)}).$$

1.1.4. *Airy-Green functions.* The Airy_β point process $\{a_{\beta,k}\}_{k=1}^\infty$ is the negative Dirichlet spectrum of the scalar stochastic Airy operator:

$$a_{\beta,k} := -\Lambda_{\beta,k}^{(\infty)}, \quad k \geq 1, \quad (1.12)$$

arranged in decreasing order.

For $\beta \in \{1, 2\}$ and a vector $\Theta = (\theta_1, \dots, \theta_q)$, $q \geq 1$, we write

$$a_{\beta,j}^\Theta := -\Lambda_{\beta,j}^{(\theta_1, \dots, \theta_q)}, \quad j \geq 1,$$

for the negative spectrum of the q -component multivariate stochastic Airy operator with boundary parameters $\theta_1, \dots, \theta_q$, arranged in decreasing order. We refer to this as the higher-rank Airy_β point process depending on Θ for $\beta = 1, 2$ [19, Remark 4.10]. For notational convention, we also allow $\Theta = ()$, and then let $a_{\beta,j}^\Theta = a_{\beta,j}$ defined in (1.12).

We now recall the Airy-Green function defined in [19]. For $\beta = 1$, the construction associated with the Airy_1 point process is given in [19, Theorem 8.1]; for general $\beta > 0$, the corresponding extension associated with the Airy_β point process is indicated in [19, Remark 4.5, Equation (4.5)] and we state its definition in the following proposition.

Proposition 1.5. *Let $\{a_{\beta,j}\}_{j \geq 1}$ be the Airy_β point process, arranged in decreasing order, and let $\{\xi_{\beta,j}\}_{j \geq 1}$ be independent and identically distributed (i.i.d.) random variables, independent of $\{a_{\beta,j}\}_{j \geq 1}$, with $\xi_{\beta,j}^2 \sim \chi_\beta^2$. Then, almost surely, the Airy-Green function for general $\beta > 0$*

$$\mathcal{G}^{(\beta)}(w) := \lim_{x \rightarrow -\infty} \left[\sum_{j: a_{\beta,j} > x} \frac{\beta^{-1} \xi_{\beta,j}^2}{w - a_{\beta,j}} - \frac{2}{\pi} \sqrt{-x} \right] \quad (1.13)$$

exists for every $w \in \mathbb{C} \setminus \{a_{\beta,j} : j \geq 1\}$, uniformly on compact subsets avoiding the poles.

In our paper, we also define the Airy-Green function associated with the higher-rank Airy_β process associated with the boundary parameters $\Theta = (\theta_1, \dots, \theta_q)$ with $\beta = 1, 2$.

Proposition 1.6. *Let $\beta \in \{1, 2\}$ and $\{\xi_{\beta,j}\}_{j \geq 1}$ be i.i.d. random variables, independent of $\{a_{\beta,j}^\Theta\}_{j \geq 1}$, with $\xi_{\beta,j}^2 \sim \chi_\beta^2$. Define the Airy-Green function*

$$\mathcal{G}_\Theta^{(\beta)}(w) := \lim_{x \rightarrow -\infty} \left[\sum_{j: a_{\beta,j}^\Theta > x} \frac{\beta^{-1} \xi_{\beta,j}^2}{w - a_{\beta,j}^\Theta} - \frac{2}{\pi} \sqrt{-x} \right]. \quad (1.14)$$

For every fixed finite vector Θ , the limit in (1.14) exists almost surely for every

$$w \in \mathbb{C} \setminus \{a_{\beta,j}^\Theta : j \geq 1\},$$

uniformly on compact subsets avoiding the poles.

For notational convention, we also allow $\Theta = ()$, and then let $\mathcal{G}_\Theta^{(\beta)}(w) = \mathcal{G}^{(\beta)}(w)$ defined in (1.13).

For later use, we record the following asymptotic behavior of the Airy Green functions at infinity in the right half plane.

Lemma 1.7. *Let $\mathcal{G}_\star$ denote either $\mathcal{G}^{(\beta)}$ defined in (1.13), or $\mathcal{G}_\Theta^{(\beta)}$ defined in (1.14).*

(1) Almost surely,

$$\lim_{\substack{|w| \rightarrow \infty \\ \Re w \geq 0}} |\mathcal{G}_*(w) + \sqrt{w}| = 0,$$

where the square root is the principal branch, positive on $(0, \infty)$. Consequently,

$$\mathcal{G}_*(x) \longrightarrow -\infty, \quad x \rightarrow +\infty.$$

(2) If, in addition, $\beta \in \{1, 2\}$, then, along the positive real axis,

$$\begin{pmatrix} x^{1/4}(\mathcal{G}_*(x) + \sqrt{x}) \\ x^{1/4} \left(-\frac{1}{\mathcal{G}'_*(x)} - 2\sqrt{x} \right) \end{pmatrix} \Rightarrow \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \frac{1}{\beta} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \right).$$

Moreover, almost surely,

$$\mathcal{G}'_*(x) = -\frac{1}{2\sqrt{x}} + o(x^{-1}), \quad \mathcal{G}''_*(x) = \frac{1}{4x^{3/2}} + o(x^{-2}).$$

We note that Proposition 1.5 is an extension of [19, Theorem 8.1] from $\beta = 1$ to $\beta > 0$, and the $\mathcal{G}_* = \mathcal{G}^{(1)}$ case of Part 1 of Lemma 1.7 is obtained in [19, Proposition 8.3].

1.2. Main results. On the fluctuation level, the critical BBP transition parameters for the spiked GOE/GUE are

$$d_i = 1 - \theta_i N^{-1/3}, \quad \theta_i \in \mathbb{R}, \quad i = 1, \dots, r, \quad (1.15)$$

and the corresponding spiked model is denoted by

$$H_N^{(D)} = H_N + \sum_{i=1}^r d_i \mathbf{w}_i \mathbf{w}_i^\top \quad \text{or} \quad H_N^{(D)} = H_N + \sum_{i=1}^r d_i \mathbf{w}_i \mathbf{w}_i^*, \quad (1.16)$$

as described in (1.8). In the rank-one tridiagonal G β E setting, we consider

$$H_N^{(\beta)} + d e_1 e_1^\top, \quad d = 1 - \theta N^{-1/3}, \quad \theta \in \mathbb{R}. \quad (1.17)$$

We now state our main results regarding the eigenvector overlap distribution in the critical BBP transition regime. The first result concerns general rank r deformation for GOE and GUE respectively. The second result gives the extension to the G β E in the rank-one case. In all cases, the overlap scale is $N^{-1/3}$. After a normalization, the limiting law is expressed in the negative reciprocal of the derivative of the corresponding Airy-Green function, evaluated at the appropriate soft-edge root.

Theorem 1.8 (GOE and GUE with finite-rank critical deformation). *Let H_N be the GOE (resp. GUE) matrix of dimension N specified by (1.1) (resp. (1.2)). Fix $r \geq 1$, and let $d_1, \dots, d_r$ be specified in (1.15), $\mathbf{w}_1, \dots, \mathbf{w}_r$ be deterministic orthonormal vectors, and the spiked model $H_N^{(D)}$ be the $N \times N$ random matrix defined in (1.16).*

Let

$$\lambda_1^{(D)} \geq \lambda_2^{(D)} \geq \dots \geq \lambda_N^{(D)}$$

be the eigenvalues of $H_N^{(D)}$, and let $\mathbf{x}_k^{(D)}$ be a unit eigenvector associated with $\lambda_k^{(D)}$. Fix $i \in \{1, \dots, r\}$ and let $k \geq 1$ be independent of N . Let $\beta = 1$ for the GOE case and $\beta = 2$ for the GUE case. Put

$$\Theta^{(i)} = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_r), \quad \mathcal{G}_{(r-1),ic}^{(\beta)} := \mathcal{G}_{\Theta^{(i)}}^{(\beta)}, \quad (1.18)$$

where $\mathcal{G}_{\Theta^{(i)}}^{(\beta)}$ is the Airy-Green function from Proposition 1.6. Let $s_{\beta,k}^{(i)}$ be the unique solution in

$$(a_{\beta,k}^{(\Theta^{(i)})}, a_{\beta,k-1}^{(\Theta^{(i)})}), \quad a_{\beta,0}^{(\Theta^{(i)})} := +\infty,$$

of

$$\mathcal{G}_{(r-1),ic}^{(\beta)}(x) = \theta_i. \quad (1.19)$$

Then, as $N \rightarrow \infty$, we have the joint convergence in distribution

$$N^{2/3}(\lambda_k^{(D)} - 2) \Rightarrow s_{\beta,k}^{(i)}, \quad N^{1/3}|\langle \mathbf{w}_i, \mathbf{x}_k^{(D)} \rangle|^2 \Rightarrow -\frac{1}{(\mathcal{G}_{(r-1),ic}^{(\beta)})'(s_{\beta,k}^{(i)})}. \quad (1.20)$$

Moreover, for any finite K , (1.20) holds jointly for $k = 1, 2, \dots, K$.

In the rank-one case, we have the following extension to the $G\beta E$, for all $\beta > 0$.

Theorem 1.9 ($G\beta E$ with rank-one critical deformation). *Let $H_N^{(\beta)}$ be the tridiagonal random matrix of dimension N defined in (1.5). Fix $\beta > 0$ and $\theta \in \mathbb{R}$. Let*

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$$

be the eigenvalues of $H_N^{(\beta)} + de_1e_1^\top$ defined in (1.17), and let $\mathbf{x}_k$ be a unit eigenvector associated with λ_k . Let $k \geq 1$ be independent of N . Let $s_{\beta,k}$ be the unique solution in

$$(a_{\beta,k}, a_{\beta,k-1}), \quad a_{\beta,0} := +\infty,$$

of

$$\mathcal{G}^{(\beta)}(x) = \theta, \quad (1.21)$$

where $\mathcal{G}^{(\beta)}$ is the Airy-Green function from Proposition 1.5. Then, as $N \rightarrow \infty$, we have the joint convergence in distribution

$$N^{2/3}(\lambda_k - 2) \Rightarrow s_{\beta,k}, \quad N^{1/3}|\langle e_1, \mathbf{x}_k \rangle|^2 \Rightarrow -\frac{1}{(\mathcal{G}^{(\beta)})'(s_{\beta,k})}. \quad (1.22)$$

Moreover, for any finite K , (1.22) holds jointly for $k = 1, 2, \dots, K$.

The next result shows that as the parameters θ_i 's go to $+\infty$ (resp. $-\infty$), the eigenvalue and eigenvector overlap properties transit from the critical regime to the subcritical (resp. supercritical) regime. For simplicity, we consider only the case that one θ_i goes to $\pm\infty$ if the rank of deformation is $r > 1$.

Proposition 1.10 (Limits as one $\theta_i \rightarrow \pm\infty$). *Suppose that $\beta > 0$. Let r be a fixed positive integer when $\beta \in \{1, 2\}$, and let $r = 1$ otherwise. Fix $i \in \{1, \dots, r\}$. When $\beta \in \{1, 2\}$, let $\Theta^{(i)}$ be defined as in (1.18); otherwise, set $\Theta^{(i)} = ()$.*

The symbols θ_i , $a_{\beta,k}^{\Theta^{(i)}}$, $s_{\beta,k}^{(i)}$, $\mathcal{G}_{(r-1),ic}^{(\beta)}$, and $\xi_{\beta,k}^2$ have the meanings specified in Theorem 1.8 for the GOE case with $\beta = 1$ and the GUE case with $\beta = 2$. For all other $\beta > 0$, they stand for θ , $a_{\beta,k}$, $s_{\beta,k}$, $\mathcal{G}^{(\beta)}$, and $\xi_{\beta,k}^2$, respectively, as in Theorem 1.9. The $o(1)$ terms in Parts (i)–(iii) are understood almost surely.

(i) *For every fixed $k \geq 1$, as $\theta_i \rightarrow +\infty$,*

$$s_{\beta,k}^{(i)} = a_{\beta,k}^{\Theta^{(i)}} + \frac{\beta^{-1}\xi_{\beta,k}^2}{\theta_i}(1 + o(1)), \quad -\frac{1}{(\mathcal{G}_{(r-1),ic}^{(\beta)})'(s_{\beta,k}^{(i)})} = \frac{\beta^{-1}\xi_{\beta,k}^2}{\theta_i^2}(1 + o(1)). \quad (1.23)$$

(ii) For every fixed $k \geq 2$, as $\theta_i \rightarrow -\infty$,

$$s_{\beta,k}^{(i)} = a_{\beta,k-1}^{\Theta^{(i)}} - \frac{\beta^{-1}\xi_{\beta,k-1}^2}{|\theta_i|}(1+o(1)), \quad -\frac{1}{(\mathcal{G}_{(r-1),i^c}^{(\beta)})'(s_{\beta,k}^{(i)})} = \frac{\beta^{-1}\xi_{\beta,k-1}^2}{\theta_i^2}(1+o(1)). \quad (1.24)$$

(iii) As $\theta_i \rightarrow -\infty$,

$$s_{\beta,1}^{(i)} = \theta_i^2(1+o(1)), \quad -\frac{1}{(\mathcal{G}_{(r-1),i^c}^{(\beta)})'(s_{\beta,1}^{(i)})} = 2|\theta_i| + o(1). \quad (1.25)$$

(iv) If, in addition, $\beta \in \{1, 2\}$, then, as $\theta_i \rightarrow -\infty$,

$$\frac{s_{\beta,1}^{(i)} - \theta_i^2}{2(-\theta_i)^{1/2}} \Rightarrow \mathcal{N}\left(0, \frac{1}{\beta}\right), \quad (-\theta_i)^{1/2} \left[-\frac{1}{(\mathcal{G}_{(r-1),i^c}^{(\beta)})'(s_{\beta,1}^{(i)})} + 2\theta_i \right] \Rightarrow \mathcal{N}\left(0, \frac{2}{\beta}\right). \quad (1.26)$$

Remark 1.4. Consider the finite-rank deformation of GOE/GUE:

- (1) From the two asymptotic relations in (1.23), we find that when one critical BBP transition parameter $d_i = 1 - \theta_i N^{-1/3}$ moves to the subcritical side as $\theta_i \rightarrow +\infty$, the top eigenvalues $\lambda_k^{(D)}$ behave as if the deformation were of rank $r-1$ specified by $\Theta^{(i)}$. This is consistent with the eigenvalue-sticking phenomenon for subcritical finite-rank deformations of Wigner matrices; see [31, Theorem 2.7 and Remark 2.13]. On the other hand, the overlap between $\mathbf{x}_k^{(D)}$ (the eigenvector associated with $\lambda_k^{(D)}$) and $\mathbf{w}_i$ (the deformation vector associated with d_i) vanishes, as if $\mathbf{w}_i$ did not represent a special direction. We note that if $\mathbf{v}$ and $\mathbf{w}$ are independent random vectors uniformly distributed on the unit sphere in $\mathbb{R}^N$ (resp. $\mathbb{C}^N$), then

$$N|\langle \mathbf{v}, \mathbf{w} \rangle|^2 \Rightarrow \xi_1^2 \quad \left(\text{resp. } \frac{1}{2}\xi_2^2 \right)$$

as $N \rightarrow \infty$. This $\beta^{-1}\xi_\beta^2$ limiting distribution is also observed in non-outlier eigenvector overlaps in the subcritical regime of spiked sample covariance matrices; see [14, Theorem 2.20].

- (2) From the two asymptotic relations in (1.25) and the two convergence statements in (1.26), we find that as one critical BBP transition parameter $d_i = 1 - \theta_i N^{-1/3}$ moves to the supercritical side as $\theta_i \rightarrow -\infty$, the top eigenvalue $\lambda_1^{(D)}$ moves away from the spectral edge 2 and becomes an outlier with Gaussian fluctuations, as featured in the supercritical regime [4, 38, 5]. On the other hand, the overlap between $\mathbf{x}_1^{(D)}$ (the eigenvector associated with $\lambda_1^{(D)}$) and $\mathbf{w}_i$ (the deformation vector associated with d_i) increases with $|\theta_i|$, and both exhibit Gaussian fluctuations [22, 21, 8, 9]. Moreover, the two asymptotic relations in (1.24) show that the other top eigenvalues behave as if they were the top eigenvalues of a rank- $(r-1)$ deformation specified by $\Theta^{(i)}$, and that the overlaps of their eigenvectors with $\mathbf{w}_i$ vanish, as if $\mathbf{w}_i$ were not a special direction for them.

Remark 1.5. Proposition 1.10 also applies to the rank-one deformation of $\text{G}\beta\text{E}$ for general $\beta > 0$. However, the Gaussian fluctuation result for the leading eigenvalue and its associated eigenvector overlap in the supercritical limit $\theta \rightarrow -\infty$ is established only for $\beta = 1, 2$, due to the corresponding restriction in Part 2 of Lemma 1.7.

1.3. Proof strategy and relation to previous results. The starting point of both the proofs of Theorems 1.8 and 1.9 is the following relation between eigenvectors and eigenvalues:

Lemma 1.11. *Let X be an N -dimensional real symmetric (resp. complex Hermitian) matrix with distinct eigenvalues $\mu_1 > \cdots > \mu_N$, $\mathbf{w}$ be a unit N -dimensional column vector in $\mathbb{R}^N$ (resp. $\mathbb{C}^N$), d be a real number, and $Y = X + d\mathbf{w}\mathbf{w}^\top$ (resp. $Y = X + d\mathbf{w}\mathbf{w}^*$) has distinct eigenvalues $\lambda_1 > \cdots > \lambda_N$. We assume $\{\mu_j\}_{j=1}^N \cup \{\lambda_j\}_{j=1}^N$ consists of $2N$ distinct real numbers, and for any $k = 1, \dots, N$, $\mathbf{x}_k$ is a unit eigenvector of Y associated with λ_k . Then*

$$\mathbf{w}^\top (\lambda_k I - X)^{-1} \mathbf{w} = \frac{1}{d}, \quad (1.27)$$

and

$$|\langle \mathbf{x}_k, \mathbf{w} \rangle|^2 = \frac{|\langle \mathbf{w}, (\lambda_k I - X)^{-1} \mathbf{w} \rangle|^2}{\langle \mathbf{w}, (\lambda_k I - X)^{-2} \mathbf{w} \rangle} = \frac{1}{d^2} \frac{1}{\langle \mathbf{w}, (\lambda_k I - X)^{-2} \mathbf{w} \rangle}. \quad (1.28)$$

In the setting of each of the theorems, identity (1.28) is rewritten as a differential identity that involves a finite- N version of the Airy-Green function specified by (1.13) or (1.14). Then by showing that the finite- N version of the Airy-Green function converges to the corresponding Airy-Green function as $N \rightarrow \infty$ in the sense of meromorphic convergence that is defined in Definition 2.1, we prove the theorems.

Instead of (1.28), another eigenvector-eigenvalue identity [23] is used as the starting point of [10, Theorem 2] for the eigenvector overlap in the critical regime of BBP transition for the GUE case. It is worth noting that since our paper is based on Lemma 1.11 that is different from the eigenvector-eigenvalue identity from [23] on which [10] is based, the GUE case of Theorem 1.8 and [10, Theorem 2] express the same limit in quite different forms. Although the two formulas describe the same limit law, their direct equivalence is not transparent.

1.4. Possible Extensions. The method is expected to extend to critically spiked real and complex Wishart ensembles which are also known as Laguerre orthogonal and unitary ensembles, and also the Laguerre β -ensemble with rank one spike. Extension to other spiked models, like the factor model and canonical correlation analysis that are considered in [19] is also possible, and we leave the discussion to future work.

Acknowledgment. Zhigang Bao was partially supported by Hong Kong RGC Grant GRF 16304724 and 17304225. Dong Wang and Yue Zhu were partially supported by the National Natural Science Foundation of China under grant number 12271502, and the University of Chinese Academy of Sciences start-up grant 118900M043.

2. PROOFS OF THE MAIN RESULTS

We begin by recording several standard spectral facts used repeatedly below.

Fact 2.1. We use the following fact throughout.

- (1) Let H_N be a GOE or GUE random matrix, and let $\mathbb{F} = \mathbb{R}$ in the GOE case and $\mathbb{F} = \mathbb{C}$ in the GUE case. Let P be a deterministic real symmetric or Hermitian matrix, and let $\mathbf{w} \in \mathbb{F}^N$ be a deterministic unit vector and $W = \mathbf{w}\mathbf{w}^\top$ in the GOE case and $W = \mathbf{w}\mathbf{w}^*$ in the GUE case. Then, for every $d \in \mathbb{R} \setminus \{0\}$,

$$\sigma(H_N + P) \cap \sigma(H_N + P + dW) = \emptyset \quad \text{almost surely.}$$

Moreover, $H_N + P$ has simple spectrum almost surely. These facts follow from the absolute continuity of the Gaussian entries and the standard discriminant/resultant polynomial argument; see [1, Corollary 2.5.4 and Lemma 2.5.5] and [41].

- (2) For the Dumitriu–Edelman tridiagonal $G\beta E$, the subdiagonal entries are almost surely positive. Hence $H_N^{(\beta)}$ is an irreducible Jacobi matrix, so its spectrum is almost surely simple and the first coordinate of each eigenvector is nonzero. Consequently, for every $d \in \mathbb{R} \setminus \{0\}$,

$$\sigma(H_N^{(\beta)}) \cap \sigma(H_N^{(\beta)} + d e_1 e_1^\top) = \emptyset \quad \text{almost surely.}$$

Moreover, if $\mathbf{u}_1, \dots, \mathbf{u}_N$ are the normalized eigenvectors of $H_N^{(\beta)}$, then the vector of squared first coordinates is independent of the eigenvalues and satisfies

$$(|\langle e_1, \mathbf{u}_1 \rangle|^2, \dots, |\langle e_1, \mathbf{u}_N \rangle|^2) \stackrel{d}{=} \left(\frac{\xi_1^2}{\sum_{\ell=1}^N \xi_\ell^2}, \dots, \frac{\xi_N^2}{\sum_{\ell=1}^N \xi_\ell^2} \right),$$

where $\xi_1, \dots, \xi_N$ are i.i.d. random variables with $\xi_j^2 \sim \chi_\beta^2$, independent of the eigenvalues; see [26, Theorem 2.12] and [25, Theorem 5.2.1].

In this paper, we adopt the following notion of meromorphic convergence that will be used in the sequel:

Definition 2.1. Let $\Omega \subset \mathbb{C}$ be an open set, and let $\{f_N\}_{N=1}^\infty$ and f be meromorphic functions on Ω . We write

$$f_N \xrightarrow{\text{mer}} f \quad \text{on } \Omega$$

if, for every compact set $K \subset \Omega$ containing no pole of f , there exists $N_0 = N_0(K)$ such that K contains no pole of f_N for all $N \geq N_0$, and

$$\sup_{z \in K} |f_N(z) - f(z)| \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

This notion agrees with locally uniform convergence in the spherical metric used in [19, Section 8]. The meromorphic functions considered in this paper belong to the class Ω_- of [19, Definition 8.10]; in particular, they have simple real poles, and their negatives are Herglotz–Nevanlinna functions.

By Cauchy’s integral formula, $f_N \xrightarrow{\text{mer}} f$ implies $f'_N \xrightarrow{\text{mer}} f'$ on the same domain. We shall use the following elementary consequence.

Lemma 2.2. Let $U \subset \mathbb{C}$ be an open set, let $x_0 \in U \cap \mathbb{R}$, and assume that $\{f_N\}_{N=1}^\infty \xrightarrow{\text{mer}} f$ on U . Suppose that x_0 is not a pole of f , and

$$f(x_0) = 0, \quad f'(x_0) \neq 0.$$

Then there exists $\delta > 0$ such that, for all sufficiently large N , f_N has exactly one zero z_N in the disc $\{z : |z - x_0| < \delta\}$, and $z_N \rightarrow x_0$. Moreover,

$$f'_N(z_N) \rightarrow f'(x_0).$$

If, in addition, $f_N(\bar{z}) = \overline{f_N(z)}$ on this disc, then $z_N \in \mathbb{R}$.

Proof. Choose $\delta > 0$ such that the closed disc $\overline{D}(x_0, \delta) \subset U$, contains no pole of f , and contains no zero of f except x_0 . Since x_0 is a simple zero, f has no zero on $\partial D(x_0, \delta)$. By uniform convergence on $\partial D(x_0, \delta)$, Rouché’s theorem implies that f_N and f have the same number of zeros in $D(x_0, \delta)$, counted with multiplicity, for all large N . Hence, f_N has exactly one zero z_N there. Shrinking δ

proves $z_N \rightarrow x_0$. The derivative convergence follows from Cauchy's integral formula. Finally, if $f_N(\bar{z}) = \overline{f_N(z)}$, then $\bar{z}_N$ is also a zero in the same disc; uniqueness gives $z_N = \bar{z}_N$. $\square$

2.1. Proof of the GOE case of Theorem 1.8 with rank 1. We begin with the GOE case of Theorem 1.8 with $r = 1$ and $i = 1$. To simplify the notation, we write (θ_1, d_1) as (θ, d) , and rewrite $s_{1,k}^{(1)}$, $H_N^{(D)}$, $\lambda_k^{(D)}$, and $\mathbf{x}_k^{(D)}$ as s_k , $H_N^{(d)}$, λ_k , and $\mathbf{x}_k$, respectively. We also write, with $\Theta^{(i)} = ()$

$$a_j := a_{1,j} = a_{1,j}^{\Theta^{(i)}}, \quad \mathcal{G} := \mathcal{G}^{(1)} = \mathcal{G}_{\Theta^{(i)}}^{(1)}.$$

Also due to the orthogonal invariance of H_N , we assume $\mathbf{w}_1 = e_1$ without loss of generality.

Let $\mu_1 \geq \mu_2 \geq \dots \geq \mu_N$ as in (1.3) be the eigenvalues of H_N . By Part 1 of Fact 2.1, we assume that $\{\mu_1, \dots, \mu_N\} \cup \{\lambda_1, \dots, \lambda_N\}$ consists of $2N$ distinct real numbers, which holds almost surely. From the orthogonal invariance of H_N , we have

$$H_N = U \operatorname{diag}(\mu_1, \dots, \mu_N) U^\top, \quad U = (\mathbf{u}_1, \dots, \mathbf{u}_N),$$

where U is Haar distributed and independent of $(\mu_1, \dots, \mu_N)$, and $\mathbf{u}_k$ is a unit eigenvector associated with μ_k ; see [1, Corollary 2.5.4]. Because U is Haar distributed and independent of the eigenvalues, its first row is a uniformly distributed unit vector, and there are i.i.d. standard Gaussian random variables $\xi_{N,1}, \dots, \xi_{N,N}$, independent of $(\mu_1, \dots, \mu_N)$, such that

$$(U^\top e_1)_i = \langle \mathbf{u}_i, e_1 \rangle = \frac{\xi_{N,i}}{(\sum_{j=1}^N \xi_{N,j}^2)^{1/2}}, \quad i = 1, \dots, N.$$

Accordingly, define the random meromorphic function

$$\begin{aligned} \tilde{\mathcal{G}}_N(w) &:= N^{1/3} e_1^\top ((2 + N^{-2/3}w)I - H_N)^{-1} e_1 - N^{1/3} \\ &= \frac{N}{\sum_{i=1}^N \xi_{N,i}^2} \sum_{i=1}^N \frac{\xi_{N,i}^2}{w - N^{2/3}(\mu_i - 2)} - N^{1/3}. \end{aligned}$$

Then (1.27) in Lemma 1.11 with $\mathbf{w} = e_1$ implies

$$\tilde{\mathcal{G}}_N(s_{k,N}) = N^{1/3} \left(\frac{1}{d} - 1 \right) = \frac{\theta}{d}, \quad \text{where } s_{k,N} := N^{2/3}(\lambda_k - 2),$$

and the zeros of $\tilde{\mathcal{G}}_N(w) - \theta/d$ are precisely $\{s_{k,N}\}_{k=1}^N$.

Since $d > 0$ for all sufficiently large N , the rank-one interlacing inequality gives

$$\lambda_k \in (\mu_k, \mu_{k-1}), \quad k \geq 1,$$

with the convention $\mu_0 = +\infty$. Equivalently,

$$s_{k,N} \in (N^{2/3}(\mu_k - 2), N^{2/3}(\mu_{k-1} - 2)).$$

Thus $s_{k,N}$ is the unique zero of $\tilde{\mathcal{G}}_N(w) - \theta/d$ in the k -th interval between two consecutive poles.

Differentiating $\tilde{\mathcal{G}}_N$, we obtain

$$(\tilde{\mathcal{G}}_N)'(w) = -N^{-1/3} e_1^\top ((2 + N^{-2/3}w)I - H_N)^{-2} e_1.$$

Evaluating $(\tilde{\mathcal{G}}_N)'(w)$ at $w = s_{k,N}$, we have

$$(\tilde{\mathcal{G}}_N)'(s_{k,N}) = -N^{-1/3} e_1^\top (\lambda_k I - H_N)^{-2} e_1.$$

By (1.28) in Lemma 1.11 with $\mathbf{w} = e_1$, we find

$$|\langle e_1, \mathbf{x}_k \rangle|^2 = -\frac{1}{N^{1/3}} \frac{1}{d^2 (\tilde{\mathcal{G}}_N)'(s_{k,N})}. \quad (2.1)$$

Next, write

$$\tilde{\mathcal{G}}_N(w) = c_N \mathcal{G}_N(w) + (c_N - 1)N^{1/3}, \quad c_N := \frac{N}{\sum_{i=1}^N \xi_{N,i}^2}, \quad (2.2)$$

where

$$\mathcal{G}_N(w) := \sum_{i=1}^N \frac{\xi_{N,i}^2}{w - N^{2/3}(\mu_i - 2)} - N^{1/3}$$

is the finite- N version of the Airy-Green function in the sense of [19, Definition 8.10 and Theorem 8.20], equivalently the function in (2.8) with $\beta = 1$. Since $\xi_{N,i}^2$ are independent χ_1^2 variables, we have $(c_N - 1)N^{1/3} \rightarrow 0$ almost surely. On the other hand, by [19, Theorem 8.20], or equivalently by Proposition 2.3 with $\beta = 1$, there exists a coupling under which the functions $\{\mathcal{G}_N\}_{N=1}^\infty$ are defined on the same probability space and

$$\mathcal{G}_N \xrightarrow{\text{mer}} \mathcal{G} \quad \text{almost surely,}$$

where $\mathcal{G} = \mathcal{G}^{(1)}$ is the Airy-Green function in Proposition 1.5. Hence, (2.2) implies that with respect to the same coupling,

$$\tilde{\mathcal{G}}_N \xrightarrow{\text{mer}} \mathcal{G} \quad \text{almost surely.} \quad (2.3)$$

Recall that $\mathcal{G}$ is defined by the Airy₁ point process $\{a_j\}_{j=1}^\infty$ and the random variables $\{\xi_j^2\}_{j=1}^\infty$, which are i.i.d. with χ_1^2 distribution and are independent of $\{a_j\}_{j=1}^\infty$. By [15, Lemma 2.7], the Dirichlet spectrum of the scalar stochastic Airy operator is almost surely simple. Hence,

$$a_1 > a_2 > \cdots \quad \text{almost surely.}$$

Conditioned on $\{a_j\}_{j=1}^\infty$, for every $x \in \mathbb{R} \setminus \{a_j : j \geq 1\}$,

$$\mathcal{G}'(x) = -\sum_{j=1}^\infty \frac{\xi_j^2}{(x - a_j)^2} < 0 \quad \text{almost surely.}$$

Thus, $\mathcal{G}$ is strictly decreasing on each interval between two consecutive Airy points. Moreover, for every $j \geq 2$,

$$\lim_{x \downarrow a_j} \mathcal{G}(x) = +\infty, \quad \lim_{x \uparrow a_{j-1}} \mathcal{G}(x) = -\infty.$$

By [19, Proposition 8.3], or equivalently by the $\beta = 1$ case of Part 1 of Lemma 1.7, almost surely,

$$\mathcal{G}(x) \longrightarrow -\infty \quad \text{as } x \rightarrow +\infty.$$

Therefore, for every $\theta \in \mathbb{R}$, the equation

$$\mathcal{G}(x) = \theta$$

has a unique real solution in each interval

$$(a_k, a_{k-1}), \quad k \geq 1,$$

where $a_0 := +\infty$. In particular, the solutions $s_k = s_k^{(1)}$ appearing in Theorem 1.8 are well defined and satisfy

$$s_1 > s_2 > \cdots \quad \text{almost surely.}$$

Since $\theta/d \rightarrow \theta$ as $N \rightarrow \infty$, it follows from (2.3) and Lemma 2.2 that, under the coupling constructed above,

$$s_{k,N} \longrightarrow s_k, \quad (2.4)$$

$$(\tilde{\mathcal{G}}_N)'(s_{k,N}) \longrightarrow \mathcal{G}'(s_k) \quad (2.5)$$

almost surely. Moreover, since the meromorphic convergence in (2.3) holds on a single event of probability one, (2.4) and (2.5) hold simultaneously for all $k \geq 1$. Since $d \rightarrow 1$, the exact overlap identity (2.1) then implies that the convergence in (1.20) holds jointly for every finite K . This completes the proof of the GOE case of Theorem 1.8 with rank 1.

2.2. Proof of Theorem 1.9. The argument is nearly the same as the proof of the rank-one GOE case of Theorem 1.8. Although the eigenvectors of the tridiagonal model are no longer uniformly distributed, the distribution of their first coordinates is given in [25, Theorem 5.2.1], which suffices for our analysis.

Recall that

$$\lambda_1 \geq \dots \geq \lambda_N$$

are the eigenvalues of $H_N^{(\beta)} + de_1e_1^\top$, and let $\mathbf{x}_k$ be a normalized eigenvector associated with λ_k . Also, let

$$\mu_1^{(\beta)} > \dots > \mu_N^{(\beta)}$$

be the eigenvalues of $H_N^{(\beta)}$, as defined in (1.6). By Part 2 of Fact 2.1, we may work on the event of probability one on which

$$\{\mu_1^{(\beta)}, \dots, \mu_N^{(\beta)}\} \cup \{\lambda_1, \dots, \lambda_N\}$$

consists of $2N$ distinct real numbers.

Using Lemma 1.11, we obtain

$$|\langle e_1, \mathbf{x}_k \rangle|^2 = \frac{1}{d^2} \frac{1}{e_1^\top (\lambda_k I - H_N^{(\beta)})^{-2} e_1}. \quad (2.6)$$

Define

$$\tilde{\mathcal{G}}_N^{(\beta)}(w) := N^{1/3} \left(e_1^\top ((2 + N^{-2/3}w)I - H_N^{(\beta)})^{-1} e_1 - 1 \right).$$

By (1.27) in Lemma 1.11, with $\mathbf{w} = e_1$, we have

$$\tilde{\mathcal{G}}_N^{(\beta)}(s_{\beta,k,N}) = N^{1/3} \left(\frac{1}{d} - 1 \right) = \frac{\theta}{d}, \quad s_{\beta,k,N} := N^{2/3}(\lambda_k - 2).$$

The zeros of

$$\tilde{\mathcal{G}}_N^{(\beta)}(w) - \frac{\theta}{d}$$

are precisely $\{s_{\beta,k,N}\}_{k=1}^N$.

Since $d > 0$ for all sufficiently large N , rank-one interlacing gives

$$\lambda_k \in (\mu_k^{(\beta)}, \mu_{k-1}^{(\beta)}), \quad k \geq 1,$$

where $\mu_0^{(\beta)} := +\infty$. Equivalently,

$$s_{\beta,k,N} \in \left(N^{2/3}(\mu_k^{(\beta)} - 2), N^{2/3}(\mu_{k-1}^{(\beta)} - 2) \right).$$

Thus, $s_{\beta,k,N}$ is the unique zero of $\tilde{\mathcal{G}}_N^{(\beta)}(w) - \theta/d$ in the k -th interval between two consecutive poles.

Differentiating $\tilde{\mathcal{G}}_N^{(\beta)}$ and evaluating at $s_{\beta,k,N}$, we obtain

$$(\tilde{\mathcal{G}}_N^{(\beta)})'(s_{\beta,k,N}) = -N^{-1/3} e_1^\top (\lambda_k I - H_N^{(\beta)})^{-2} e_1.$$

Together with (2.6), this gives

$$|\langle e_1, \mathbf{x}_k \rangle|^2 = -\frac{N^{-1/3}}{d^2 (\tilde{\mathcal{G}}_N^{(\beta)})'(s_{\beta,k,N})}. \quad (2.7)$$

Write the spectral decomposition as

$$H_N^{(\beta)} = \sum_{i=1}^N \mu_i^{(\beta)} \mathbf{u}_{\beta,i} \mathbf{u}_{\beta,i}^\top.$$

By [25, Theorem 5.2.1], there exist i.i.d. random variables

$$\xi_{\beta,N,1}^2, \dots, \xi_{\beta,N,N}^2 \sim \chi_\beta^2,$$

independent of $\{\mu_i^{(\beta)}\}_{i=1}^N$, such that

$$\begin{aligned} & (|\langle e_1, \mathbf{u}_{\beta,1} \rangle|^2, \dots, |\langle e_1, \mathbf{u}_{\beta,N} \rangle|^2) \\ & \stackrel{d}{=} \left(\frac{\beta^{-1} \xi_{\beta,N,1}^2}{\sum_{j=1}^N \beta^{-1} \xi_{\beta,N,j}^2}, \dots, \frac{\beta^{-1} \xi_{\beta,N,N}^2}{\sum_{j=1}^N \beta^{-1} \xi_{\beta,N,j}^2} \right). \end{aligned}$$

Define

$$\mathcal{G}_N^{(\beta)}(w) := \sum_{i=1}^N \frac{\beta^{-1} \xi_{\beta,N,i}^2}{w - N^{2/3}(\mu_i^{(\beta)} - 2)} - N^{1/3}. \quad (2.8)$$

Then, analogously to (2.2),

$$\tilde{\mathcal{G}}_N^{(\beta)}(w) = c_{\beta,N} \mathcal{G}_N^{(\beta)}(w) + (c_{\beta,N} - 1)N^{1/3}, \quad c_{\beta,N} := \frac{N}{\sum_{i=1}^N \beta^{-1} \xi_{\beta,N,i}^2}.$$

Moreover,

$$(c_{\beta,N} - 1)N^{1/3} \longrightarrow 0 \quad \text{almost surely.}$$

We use the following extension of [19, Theorem 8.20] from $\beta = 1$ to all $\beta > 0$.

Proposition 2.3. *Suppose that $\{\xi_{\beta,N,i}\}_{i=1}^N$ are independent of $\{\mu_i^{(\beta)}\}_{i=1}^N$ and satisfy $\xi_{\beta,N,i}^2 \sim \chi_\beta^2$. There exists a coupling that places*

$$\left(\{\mu_i^{(\beta)}\}_{i=1}^N, \{\xi_{\beta,N,i}^2\}_{i=1}^N \right)_{N=1}^\infty$$

and $\mathcal{G}^{(\beta)}$ on the same probability space such that, almost surely,

$$\mathcal{G}_N^{(\beta)} \xrightarrow{\text{mer}} \mathcal{G}^{(\beta)}. \quad (2.9)$$

The proof of Proposition 2.3 is given in Section 4.2.

Recall that $\mathcal{G}^{(\beta)}$ is defined in [19, Remark 4.5] and (1.13) by the Airy $_\beta$ point process $\{a_{\beta,j}\}_{j=1}^\infty$ and the random variables $\{\xi_{\beta,j}^2\}_{j=1}^\infty$, which are i.i.d. with χ_β^2 distribution and are independent of $\{a_{\beta,j}\}_{j=1}^\infty$.

By [15, Lemma 2.7], the Dirichlet spectrum of the scalar stochastic Airy operator is almost surely simple. Hence,

$$a_{\beta,1} > a_{\beta,2} > \dots \quad \text{almost surely.}$$

Conditioned on $\{a_{\beta,j}\}_{j=1}^\infty$, for every $x \in \mathbb{R} \setminus \{a_{\beta,j} : j \geq 1\}$,

$$(\mathcal{G}^{(\beta)})'(x) = - \sum_{j=1}^{\infty} \frac{\beta^{-1} \xi_{\beta,j}^2}{(x - a_{\beta,j})^2} < 0 \quad \text{almost surely.}$$

Thus, $\mathcal{G}^{(\beta)}$ is strictly decreasing on each interval between two consecutive Airy_β points. Moreover, for every $j \geq 2$,

$$\lim_{x \downarrow a_{\beta,j}} \mathcal{G}^{(\beta)}(x) = +\infty, \quad \lim_{x \uparrow a_{\beta,j-1}} \mathcal{G}^{(\beta)}(x) = -\infty.$$

By Part 1 of Lemma 1.7, almost surely,

$$\mathcal{G}^{(\beta)}(x) \longrightarrow -\infty \quad \text{as } x \rightarrow +\infty.$$

Therefore, for every $\theta \in \mathbb{R}$, the equation

$$\mathcal{G}^{(\beta)}(x) = \theta$$

has a unique real solution

$$s_{\beta,k} \in (a_{\beta,k}, a_{\beta,k-1}), \quad k \geq 1,$$

where $a_{\beta,0} := +\infty$.

Since $\theta/d \rightarrow \theta$ as $N \rightarrow \infty$, it follows from (2.9) and Lemma 2.2 that

$$s_{\beta,k,N} \longrightarrow s_{\beta,k}, \quad (2.10)$$

$$(\tilde{\mathcal{G}}_N^{(\beta)})'(s_{\beta,k,N}) \longrightarrow (\mathcal{G}^{(\beta)})'(s_{\beta,k}) \quad (2.11)$$

almost surely under the coupling above. Moreover, since the meromorphic convergence holds on a single event of probability one, (2.10) and (2.11) hold simultaneously for all $k \geq 1$. Since $d \rightarrow 1$, the exact overlap identity (2.7) then implies that the convergence in (1.22) holds jointly for every finite K . This completes the proof of Theorem 1.9.

2.3. Proof of the GOE case of Theorem 1.8 by induction on the rank. We prove by induction on r the overlap convergence in Theorem 1.8, simultaneously with the almost-sure simplicity of the higher-rank Airy_1 point process

$$\{a_{1,k}^\Theta\}_{k \geq 1}, \quad \Theta = (\theta_1, \dots, \theta_r),$$

that is,

$$a_{1,1}^\Theta > a_{1,2}^\Theta > \dots \quad \text{almost surely.}$$

For $r = 1$, the point process $\{a_{1,k}^{(\theta_1)}\}_{k \geq 1}$ is the negative spectrum of the scalar stochastic Airy operator with Robin boundary condition $f'(0) = \theta_1 f(0)$. By [15, Lemma 2.7], every eigenspace of this operator is at most one-dimensional. Therefore,

$$a_{1,1}^{(\theta_1)} > a_{1,2}^{(\theta_1)} > \dots \quad \text{almost surely.}$$

The rank-one overlap convergence was proved in Section 2.1, so the induction base is complete.

Assume now that both assertions hold for all ranks up to $r-1$, and consider rank r . By orthogonal invariance and relabeling the spikes, we may assume

$$\mathbf{w}_j = e_j, \quad 1 \leq j \leq r, \quad \text{and} \quad i = r.$$

For notational simplicity, set

$$\begin{aligned} \Theta^{(r)} &:= (\theta_1, \dots, \theta_{r-1}), & a_k^{\Theta^{(r)}} &:= a_{1,k}^{\Theta^{(r)}}, \\ s_k^{(r)} &:= s_{1,k}^{(r)}, & \mathcal{G}_{r-1} &:= \mathcal{G}_{(r-1),r^c}^{(1)}. \end{aligned} \quad (2.12)$$

By the induction hypothesis, the rank- $(r-1)$ Airy point process is almost surely simple:

$$a_1^{\Theta^{(r)}} > a_2^{\Theta^{(r)}} > \dots \quad \text{almost surely.}$$

Define

$$\hat{H}_N := H_N + \sum_{j=1}^{r-1} d_j e_j e_j^\top, \quad H_N^{(D)} = \hat{H}_N + d_r e_r e_r^\top.$$

Let

$$\hat{\mu}_{1,N} \geq \dots \geq \hat{\mu}_{N,N}$$

be the eigenvalues of $\hat{H}_N$, with corresponding normalized eigenvectors $\hat{\mathbf{u}}_{1,N}, \dots, \hat{\mathbf{u}}_{N,N}$. By Part 1 of Fact 2.1, the spectra of $\hat{H}_N$ and $H_N^{(D)}$ are almost surely simple and disjoint. The choices of signs of the eigenvectors are immaterial, since only their squared coordinates are used below.

Put

$$a_{m,N} := N^{2/3}(\hat{\mu}_{m,N} - 2)$$

and define

$$\begin{aligned} \tilde{\mathcal{G}}_{N,r}(w) &:= N^{1/3} \left[e_r^\top \left((2 + N^{-2/3}w)I - \hat{H}_N \right)^{-1} e_r - 1 \right] \\ &= \sum_{m=1}^N \frac{N |\langle e_r, \hat{\mathbf{u}}_{m,N} \rangle|^2}{w - a_{m,N}} - N^{1/3}. \end{aligned} \tag{2.13}$$

By (1.27) in Lemma 1.11,

$$\tilde{\mathcal{G}}_{N,r}(s_{k,N}^{(r)}) = \frac{\theta_r}{d_r}, \quad s_{k,N}^{(r)} := N^{2/3}(\lambda_k^{(D)} - 2).$$

Since $d_r > 0$ for all sufficiently large N , rank-one interlacing gives

$$s_{k,N}^{(r)} \in (a_{k,N}, a_{k-1,N}), \quad a_{0,N} := +\infty.$$

Thus, $s_{k,N}^{(r)}$ is the unique zero of $\tilde{\mathcal{G}}_{N,r} - \theta_r/d_r$ in this interval.

The exact overlap identity in Lemma 1.11 gives

$$|\langle e_r, \mathbf{x}_k^{(D)} \rangle|^2 = -\frac{N^{-1/3}}{d_r^2 \tilde{\mathcal{G}}'_{N,r}(s_{k,N}^{(r)})}. \tag{2.14}$$

We use the following joint edge convergence.

Lemma 2.4. *For every fixed $M \geq 1$,*

$$\left((a_{m,N})_{m=1}^M, (N |\langle e_r, \hat{\mathbf{u}}_{m,N} \rangle|^2)_{m=1}^M \right) \implies \left((a_m^{\Theta^{(r)}})_{m=1}^M, (\eta_m^2)_{m=1}^M \right), \tag{2.15}$$

where $\eta_1, \dots, \eta_M$ are i.i.d. standard Gaussian random variables, independent of $\{a_m^{\Theta^{(r)}}\}_{m \geq 1}$.

As a consequence, we obtain the higher-rank analogue of [19, Theorem 8.20].

Proposition 2.5. *There exists a coupling under which $\tilde{\mathcal{G}}_{N,r}$ and $\mathcal{G}_{r-1}$ are defined on the same probability space and*

$$\tilde{\mathcal{G}}_{N,r} \xrightarrow{\text{mer}} \mathcal{G}_{r-1} \quad \text{almost surely.}$$

The proof of Proposition 2.5 is given in Section 4.4. We first complete the induction argument and then prove Lemma 2.4.

Recall that $\mathcal{G}_{r-1}$ is the Airy–Green function defined from the rank- $(r-1)$ Airy point process

$$\{a_k^{\Theta^{(r)}}\}_{k \geq 1}$$

and an independent sequence $\{\xi_k^2\}_{k \geq 1}$ of χ_1^2 random variables. Conditioned on the Airy points,

$$\mathcal{G}'_{r-1}(x) = - \sum_{j=1}^{\infty} \frac{\xi_j^2}{(x - a_j^{\Theta^{(r)}})^2} < 0$$

for every $x \notin \{a_j^{\Theta^{(r)}} : j \geq 1\}$, almost surely. Moreover, for every $j \geq 2$,

$$\lim_{x \downarrow a_j^{\Theta^{(r)}}} \mathcal{G}_{r-1}(x) = +\infty, \quad \lim_{x \uparrow a_{j-1}^{\Theta^{(r)}}} \mathcal{G}_{r-1}(x) = -\infty,$$

and Part 1 of Lemma 1.7 gives

$$\mathcal{G}_{r-1}(x) \longrightarrow -\infty \quad \text{as } x \rightarrow +\infty.$$

Consequently, for every $k \geq 1$, the equation

$$\mathcal{G}_{r-1}(x) = \theta_r$$

has a unique solution

$$s_k^{(r)} \in (a_k^{\Theta^{(r)}}, a_{k-1}^{\Theta^{(r)}}), \quad a_0^{\Theta^{(r)}} := +\infty.$$

In particular,

$$s_1^{(r)} > s_2^{(r)} > \dots \quad \text{almost surely.}$$

Since $\theta_r/d_r \rightarrow \theta_r$, Proposition 2.5 and Lemma 2.2 imply, under the coupling above,

$$\left(s_{k,N}^{(r)}, \tilde{\mathcal{G}}'_{N,r}(s_{k,N}^{(r)})\right) \longrightarrow \left(s_k^{(r)}, \mathcal{G}'_{r-1}(s_k^{(r)})\right) \quad (2.16)$$

almost surely, simultaneously for all $k \geq 1$.

On the other hand, Proposition 1.4 gives, for every fixed $K \geq 1$,

$$(s_{k,N}^{(r)})_{k=1}^K \Longrightarrow (a_{1,k}^{\Theta})_{k=1}^K.$$

Combining this with (2.16) and uniqueness of the weak limit yields

$$(s_k^{(r)})_{k=1}^K \stackrel{d}{=} (a_{1,k}^{\Theta})_{k=1}^K. \quad (2.17)$$

Since the roots $\{s_k^{(r)}\}_{k \geq 1}$ are almost surely strictly ordered, and since K is arbitrary, (2.17) proves that the rank- r Airy point process is almost surely simple:

$$a_{1,1}^{\Theta} > a_{1,2}^{\Theta} > \dots \quad \text{almost surely.}$$

This completes the induction proof of the simplicity assertion in Proposition 1.3 for $\beta = 1$.

Finally, by (2.14), for every fixed $K \geq 1$,

$$\left(s_{k,N}^{(r)}, N^{1/3} |\langle e_r, \mathbf{x}_k^{(D)} \rangle|^2\right)_{k=1}^K \longrightarrow \left(s_k^{(r)}, -\frac{1}{\mathcal{G}'_{r-1}(s_k^{(r)})}\right)_{k=1}^K \quad \text{almost surely}$$

under the coupling above. Hence the corresponding convergence holds jointly in distribution. This completes the simultaneous induction and the proof of the GOE case of Theorem 1.8.

Proof of Lemma 2.4. By Proposition 1.4, for every fixed $M \geq 1$,

$$(a_{m,N})_{m=1}^M \implies (a_m^{\Theta^{(r)}})_{m=1}^M.$$

Define

$$A_N := (\langle e_i, \hat{\mathbf{u}}_{j,N} \rangle)_{1 \leq i \leq r-1, 1 \leq j \leq M} \in \mathbb{R}^{(r-1) \times M},$$

and

$$B_N := (\langle e_i, \hat{\mathbf{u}}_{j,N} \rangle)_{r \leq i \leq N, 1 \leq j \leq M} \in \mathbb{R}^{(N-r+1) \times M}.$$

The choices of signs of the eigenvectors only multiply the columns of A_N and B_N by signs and hence do not affect the squared coordinates considered below.

Since $\hat{\mathbf{u}}_{1,N}, \dots, \hat{\mathbf{u}}_{M,N}$ are orthonormal,

$$A_N^\top A_N + B_N^\top B_N = I_M.$$

Set

$$C_N := B_N^\top B_N = I_M - A_N^\top A_N.$$

The law of $\hat{H}_N$ is invariant under conjugation by matrices of the form

$$I_{r-1} \oplus O, \quad O \in \mathcal{O}(N-r+1).$$

Consequently, conditionally on the eigenvalues and on the upper block A_N , the lower block B_N , modulo the irrelevant column signs, is Haar distributed on the orthogonal orbit determined by $B_N^\top B_N = C_N$. Therefore, for the sign-invariant quantities considered here, its conditional law is the same as that of

$$Q_N C_N^{1/2},$$

where Q_N is Haar distributed on the Stiefel manifold

$$V_M(\mathbb{R}^{N-r+1}) = \left\{ Q \in \mathbb{R}^{(N-r+1) \times M} : Q^\top Q = I_M \right\}$$

and may be taken independent of the conditioning variables.

By the induction hypothesis, for every fixed $1 \leq i \leq r-1$ and $1 \leq j \leq M$, the sequence

$$N^{1/3} |\langle e_i, \hat{\mathbf{u}}_{j,N} \rangle|^2$$

is tight. Since r and M are fixed,

$$\|A_N\|_{\text{HS}}^2 = \sum_{i=1}^{r-1} \sum_{j=1}^M |\langle e_i, \hat{\mathbf{u}}_{j,N} \rangle|^2 = O_{\mathbb{P}}(N^{-1/3}).$$

It follows that

$$C_N^{1/2} = I_M + O_{\mathbb{P}}(N^{-1/3})$$

in operator norm.

Let q_N denote the first row of Q_N . For fixed M ,

$$\sqrt{N-r+1} q_N \implies (\eta_1, \dots, \eta_M),$$

where $\eta_1, \dots, \eta_M$ are i.i.d. standard Gaussian random variables; see [34, Theorem 2.8]. Since Q_N is independent of the conditioning variables, the limiting Gaussian vector is independent of every subsequential limit of

$$((a_{m,N})_{m=1}^M, A_N).$$

Combining these observations with $(N-r+1)/N \rightarrow 1$, we obtain (2.15) which proves the lemma. $\square$

2.4. Sketch of proof of the GUE case of Theorem 1.8. The proof is the complex analogue of the GOE argument. We prove simultaneously the overlap convergence and the almost-sure simplicity of the higher-rank Airy_2 point process $\{a_{2,k}^{\Theta}\}_{k \geq 1}$ ($\Theta = (\theta_1, \dots, \theta_r)$).

For $r = 1$, this point process is the negative spectrum of the scalar stochastic Airy operator with Robin boundary condition $f'(0) = \theta_1 f(0)$. Its almost-sure simplicity follows directly from [15, Lemma 2.7]. The rank-one overlap convergence follows from the $\beta = 2$ case of Theorem 1.9. This establishes the induction base.

Assume now that the rank- $(r-1)$ Airy_2 point process is almost surely simple. By unitary invariance and relabeling the spikes, we may assume that $\mathbf{w}_j = e_j$, $1 \leq j \leq r$, and $i = r$. Set $\Theta^{(r)} = (\theta_1, \dots, \theta_{r-1})$ and $\mathcal{G}_{r-1}^{(2)} := \mathcal{G}_{(r-1),rc}^{(2)}$. Thus,

$$a_{2,1}^{\Theta^{(r)}} > a_{2,2}^{\Theta^{(r)}} > \dots \quad \text{almost surely.}$$

Define

$$\widehat{H}_N := H_N + \sum_{j=1}^{r-1} d_j e_j e_j^*, \quad H_N^{(D)} = \widehat{H}_N + d_r e_r e_r^*.$$

Let $\widehat{\mu}_{1,N} \geq \dots \geq \widehat{\mu}_{N,N}$ be the eigenvalues of $\widehat{H}_N$, with corresponding normalized eigenvectors $\widehat{\mathbf{u}}_{1,N}, \dots, \widehat{\mathbf{u}}_{N,N}$, and put $a_{2,m,N} := N^{2/3}(\widehat{\mu}_{m,N} - 2)$. Define

$$\begin{aligned} \widetilde{\mathcal{G}}_{N,r}^{(2)}(w) &:= N^{1/3} \left[e_r^* ((2 + N^{-2/3}w)I - \widehat{H}_N)^{-1} e_r - 1 \right] \\ &= \sum_{m=1}^N \frac{N |\langle e_r, \widehat{\mathbf{u}}_{m,N} \rangle|^2}{w - a_{2,m,N}} - N^{1/3}. \end{aligned}$$

Set $s_{2,k,N}^{(r)} := N^{2/3}(\lambda_k^{(D)} - 2)$. The rank-one determinant identity and Lemma 1.11, with transpose replaced by adjoint, give

$$\widetilde{\mathcal{G}}_{N,r}^{(2)}(s_{2,k,N}^{(r)}) = \frac{\theta_r}{d_r}, \quad |\langle e_r, \mathbf{x}_k^{(D)} \rangle|^2 = -\frac{N^{-1/3}}{d_r^2 (\widetilde{\mathcal{G}}_{N,r}^{(2)})'(s_{2,k,N}^{(r)})}. \quad (2.18)$$

The proof of Lemma 2.4 carries over to the complex setting after replacing $\mathbb{R}$, $\text{O}(\cdot)$, and $(\cdot)^\top$ by $\mathbb{C}$, $\text{U}(\cdot)$, and $(\cdot)^*$, respectively. In particular, for every fixed $M \geq 1$,

$$\left((a_{2,m,N})_{m=1}^M, (N |\langle e_r, \widehat{\mathbf{u}}_{m,N} \rangle|^2)_{m=1}^M \right) \implies \left((a_{2,m}^{\Theta^{(r)}})_{m=1}^M, (|\eta_m|^2)_{m=1}^M \right),$$

where $\eta_1, \dots, \eta_M$ are i.i.d. standard complex Gaussian random variables, independent of $\{a_{2,m}^{\Theta^{(r)}}\}_{m \geq 1}$.

Here $|\eta_m|^2 \stackrel{d}{=} \frac{1}{2} \chi_2^2$.

Indeed, the conditional Haar-orbit argument in the proof of Lemma 2.4 is unchanged, with the real Stiefel manifold replaced by its complex counterpart. The necessary second and fourth moment estimates follow from Lemma 4.3 with $\beta_F = 2$.

The proof of Proposition 2.5 therefore also carries over to the complex setting and yields a coupling under which

$$\widetilde{\mathcal{G}}_{N,r}^{(2)} \xrightarrow{\text{mer}} \mathcal{G}_{r-1}^{(2)} \quad \text{almost surely.} \quad (2.19)$$

The limiting weights are precisely $\frac{1}{2} \chi_2^2$, so the limiting function is $\mathcal{G}_{(r-1),rc}^{(2)}$.

By the induction hypothesis, the poles $\{a_{2,k}^{\Theta^{(r)}}\}_{k \geq 1}$ are almost surely strictly ordered. The same argument as in the GOE case shows that, for every $k \geq 1$, the equation $\mathcal{G}_{r-1}^{(2)}(x) = \theta_r$ has a unique solution

$$s_{2,k}^{(r)} \in (a_{2,k}^{\Theta^{(r)}}, a_{2,k-1}^{\Theta^{(r)}}), \quad a_{2,0}^{\Theta^{(r)}} := +\infty.$$

In particular,

$$s_{2,1}^{(r)} > s_{2,2}^{(r)} > \dots \quad \text{almost surely.}$$

Since $\theta_r/d_r \rightarrow \theta_r$, (2.19) and Lemma 2.2 imply

$$(s_{2,k,N}^{(r)}, (\tilde{\mathcal{G}}_{N,r}^{(2)})'(s_{2,k,N}^{(r)})) \longrightarrow (s_{2,k}^{(r)}, (\mathcal{G}_{r-1}^{(2)})'(s_{2,k}^{(r)})) \quad (2.20)$$

almost surely under the corresponding coupling, simultaneously for all $k \geq 1$.

For every fixed $K \geq 1$, Proposition 1.4 gives

$$(s_{2,k,N}^{(r)})_{k=1}^K \Longrightarrow (a_{2,k}^{\Theta})_{k=1}^K.$$

Combining this convergence with (2.20) and uniqueness of the weak limit yields

$$(s_{2,k}^{(r)})_{k=1}^K \stackrel{d}{=} (a_{2,k}^{\Theta})_{k=1}^K.$$

Since the roots on the left are almost surely strictly ordered and K is arbitrary, this proves the almost-sure simplicity of the rank- r Airy₂ point process:

$$a_{2,1}^{\Theta} > a_{2,2}^{\Theta} > \dots \quad \text{almost surely.}$$

This establishes the simplicity assertion in Proposition 1.3 for $\beta = 2$.

Finally, substituting (2.20) into the exact overlap identity (2.18) gives, for every fixed $K \geq 1$,

$$\left(s_{2,k,N}^{(r)}, N^{1/3} |\langle e_r, \mathbf{x}_k^{(D)} \rangle|^2\right)_{k=1}^K \longrightarrow \left(s_{2,k}^{(r)}, -\frac{1}{(\mathcal{G}_{r-1}^{(2)})'(s_{2,k}^{(r)})}\right)_{k=1}^K \quad \text{almost surely}$$

under the coupling above. Hence the corresponding convergence holds jointly in distribution. This completes the simultaneous induction and the proof of the GUE case of Theorem 1.8.

3. PROOF OF PROPOSITION 1.10

In this section, we use the notation of Proposition 1.10. When $\beta \notin \{1, 2\}$, we have $r = 1$ and $\Theta^{(i)} = ()$, and the symbols

$$\theta_i, \quad a_{\beta,k}^{\Theta^{(i)}}, \quad s_{\beta,k}^{(i)}, \quad \mathcal{G}_{(r-1),ic}^{(\beta)}, \quad \xi_{\beta,k}^2$$

are understood as

$$\theta, \quad a_{\beta,k}, \quad s_{\beta,k}, \quad \mathcal{G}^{(\beta)}, \quad \xi_{\beta,k}^2,$$

respectively, as in Theorem 1.9.

Near a pole $a_{\beta,m}^{\Theta^{(i)}}$, Proposition 1.6 for $\beta \in \{1, 2\}$, and Proposition 1.5 for general $\beta > 0$, give

$$\mathcal{G}_{(r-1),ic}^{(\beta)}(x) = \frac{\beta^{-1} \xi_{\beta,m}^2}{x - a_{\beta,m}^{\Theta^{(i)}}} + O(1), \quad (3.1)$$

$$(\mathcal{G}_{(r-1),ic}^{(\beta)})'(x) = -\frac{\beta^{-1} \xi_{\beta,m}^2}{(x - a_{\beta,m}^{\Theta^{(i)}})^2} + O(1). \quad (3.2)$$

Moreover, $(\mathcal{G}_{(r-1),ic}^{(\beta)})'(x) < 0$ away from the poles, so $\mathcal{G}_{(r-1),ic}^{(\beta)}$ is strictly decreasing on every interval between consecutive poles.

As $\theta_i \rightarrow +\infty$, monotonicity gives

$$s_{\beta,k}^{(i)} \downarrow a_{\beta,k}^{\Theta^{(i)}}.$$

Taking $m = k$ in (3.1), we obtain

$$\theta_i = \frac{\beta^{-1} \xi_{\beta,k}^2}{s_{\beta,k}^{(i)} - a_{\beta,k}^{\Theta^{(i)}}} + O(1),$$

which proves the first asymptotic relation in (1.23). Substitution into (3.2) then proves the second asymptotic relation in (1.23).

If $\theta_i \rightarrow -\infty$ and $k \geq 2$, then

$$s_{\beta,k}^{(i)} \uparrow a_{\beta,k-1}^{\Theta^{(i)}}.$$

Taking $m = k - 1$ in (3.1) gives

$$\theta_i = \frac{\beta^{-1} \xi_{\beta,k-1}^2}{s_{\beta,k}^{(i)} - a_{\beta,k-1}^{\Theta^{(i)}}} + O(1).$$

This proves the first asymptotic relation in (1.24); the second asymptotic relation follows from (3.2).

It remains to consider $k = 1$ as $\theta_i \rightarrow -\infty$. Monotonicity gives

$$s_{\beta,1}^{(i)} \rightarrow +\infty.$$

By the defining root equations (1.19) and (1.21), together with Part 1 of Lemma 1.7,

$$\theta_i = \mathcal{G}_{(r-1),ic}^{(\beta)}(s_{\beta,1}^{(i)}) = -\sqrt{s_{\beta,1}^{(i)}} + o(1),$$

which proves the first asymptotic relation in (1.25).

The same lemma gives, uniformly in the right half-plane,

$$\mathcal{G}_{(r-1),ic}^{(\beta)}(z) + \sqrt{z} = o(1), \quad |z| \rightarrow \infty.$$

Cauchy's formula on $|z - x| = x/2$ therefore yields

$$(\mathcal{G}_{(r-1),ic}^{(\beta)})'(x) = -\frac{1}{2\sqrt{x}} + o(x^{-1}), \quad x \rightarrow +\infty.$$

Evaluating this at $x = s_{\beta,1}^{(i)}$ and using the first asymptotic relation in (1.25), we obtain

$$(\mathcal{G}_{(r-1),ic}^{(\beta)})'(s_{\beta,1}^{(i)}) = -\frac{1}{2|\theta_i|} + o(|\theta_i|^{-2}).$$

Consequently,

$$-\frac{1}{(\mathcal{G}_{(r-1),ic}^{(\beta)})'(s_{\beta,1}^{(i)})} = 2|\theta_i| + o(1),$$

which proves the second asymptotic relation in (1.25).

We now assume $\beta \in \{1, 2\}$, since the remaining argument uses Part 2 of Lemma 1.7. For notational convenience within this part only, set

$$\begin{aligned} t &:= -\theta_i, & \mathcal{G} &:= \mathcal{G}_{(r-1),ic}^{(\beta)}, & s_t &:= s_{\beta,1}^{(i)}, \\ U_t &:= t^{1/2}(\mathcal{G}(t^2) + t), & V_t &:= t^{1/2} \left(-\frac{1}{\mathcal{G}'(t^2)} - 2t \right). \end{aligned}$$

Applying Part 2 of Lemma 1.7 with $x = t^2$, we obtain

$$\begin{pmatrix} U_t \\ V_t \end{pmatrix} \Rightarrow \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \frac{1}{\beta} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}\right). \quad (3.3)$$

Since $\mathcal{G}(s_t) = -t$ and $\mathcal{G}(s_t) = -\sqrt{s_t} + o(1)$, we have $\sqrt{s_t} = t + o(1)$, and hence

$$s_t = t^2 + o(t) \quad \text{almost surely.}$$

By the mean value theorem, for some random point ζ_t between t^2 and s_t ,

$$0 = \mathcal{G}(s_t) + t = \mathcal{G}(t^2) + t + \mathcal{G}'(\zeta_t)(s_t - t^2).$$

Since $\zeta_t = t^2 + o(t)$, the derivative asymptotics in Lemma 1.7 give

$$\mathcal{G}'(\zeta_t) = -\frac{1}{2t}(1 + o(1)) \quad \text{almost surely.}$$

Together with $U_t = O_{\mathbb{P}}(1)$, this yields

$$\frac{s_t - t^2}{t^{1/2}} = 2U_t + o_{\mathbb{P}}(1). \quad (3.4)$$

Consequently,

$$\frac{s_t - t^2}{2t^{1/2}} \Rightarrow \mathcal{N}\left(0, \frac{1}{\beta}\right),$$

which proves the first convergence in (1.26).

For the overlap fluctuation, set

$$R(x) := -\frac{1}{\mathcal{G}'(x)}.$$

The derivative asymptotics in Lemma 1.7 imply

$$R'(x) = \frac{\mathcal{G}''(x)}{\mathcal{G}'(x)^2} = \frac{1}{\sqrt{x}}(1 + o(1)) \quad \text{almost surely.}$$

Applying the mean value theorem once more, for some random point η_t between t^2 and s_t ,

$$R(s_t) - R(t^2) = R'(\eta_t)(s_t - t^2).$$

Since $\eta_t = t^2 + o(t)$, it follows from (3.4) that

$$t^{1/2}(R(s_t) - R(t^2)) = 2U_t + o_{\mathbb{P}}(1).$$

Therefore,

$$\begin{aligned} t^{1/2} \left[-\frac{1}{\mathcal{G}'(s_t)} - 2t \right] &= t^{1/2}(R(s_t) - R(t^2)) + t^{1/2}(R(t^2) - 2t) \\ &= 2U_t + V_t + o_{\mathbb{P}}(1). \end{aligned}$$

If (U, V) denotes the Gaussian limit in (3.3), then

$$\text{Var}(2U + V) = \frac{4}{\beta} + \frac{2}{\beta} - \frac{4}{\beta} = \frac{2}{\beta}.$$

It follows that

$$t^{1/2} \left[-\frac{1}{\mathcal{G}'(s_t)} - 2t \right] \Rightarrow \mathcal{N}\left(0, \frac{2}{\beta}\right),$$

which proves the second convergence in (1.26) and completes the proof of Proposition 1.10.

4. PROPERTIES OF AIRY-GREEN FUNCTIONS

In this section we prove Propositions 1.5, 1.6 and Lemma 1.7 in Section 1.1, and prove Propositions 2.3 and 2.5 that are used in the proofs of Theorems 1.8 and 1.9.

4.1. Proofs of Proposition 1.5 and Lemma 1.7 for $\mathcal{G}^{(\beta)}$. The proof of Proposition 1.5 extends the argument of [19, Theorem 8.1] from $\beta = 1$ to general $\beta > 0$. For Lemma 1.7, the proof of Part 1 is adapted from the argument of [19, Proposition 8.3], whereas the fluctuation and derivative estimates in Part 2 require additional arguments.

4.1.1. Rigidity input for the Airy_β point process. In the proof of Proposition 1.5, we need the rigidity result of the Airy_β point process that is a generalization of [19, Lemma 8.9].

Lemma 4.1. *For every $\varepsilon > 0$, there exists a random variable $\mathcal{J} = \mathcal{J}(\varepsilon)$ such that, almost surely, for all $j > \mathcal{J}$,*

$$\left| a_{\beta,j} + \left(\frac{3\pi j}{2} \right)^{2/3} \right| \leq j^{\frac{1}{6} + \varepsilon}. \quad (4.1)$$

The $\beta = 1$ case of Lemma 4.1 is stated in [19, Lemma 8.9], where $a_{\beta,j}$ is denoted by $\mathfrak{a}_j$, and the exponent $1/6 + \varepsilon$ can be strengthened to ε . For $\beta = 2$, the corresponding stronger estimate follows from the mean asymptotics stated immediately before [40, Theorem 1] and the variance estimate in that theorem. In [19, Section 8], the $\beta = 1$ estimate is obtained using the corresponding mean asymptotics together with a variance estimate deduced from the $\beta = 2$ case via the Forrester–Rains relation [28, Theorem 4.3]; see also [36]. Here we instead use [2, Theorem 1 and Corollary 2]; see also [42, Proposition 3.2].

Proof. From [2, Theorem 1 and Corollary 2], we have that for every $\varepsilon \in (0, 1/6)$, there exist $k_0 = k_0(\varepsilon)$, $\chi = \chi(\varepsilon) > 0$, and $C = C(\varepsilon) > 0$ such that for all $k \geq k_0$,

$$\mathbb{P} \left(\left| a_{\beta,k} + \left(\frac{3\pi k}{2} \right)^{2/3} \right| \geq Ck^{\frac{1}{6} + \varepsilon} \right) \leq e^{-k^\chi}.$$

Then the Borel–Cantelli lemma yields the claim. $\square$

4.1.2. Proof of Proposition 1.5. Analogous to the proof of [19, Theorem 8.1], we define, like the decomposition in [19, Equations (8.19), (8.20) and (8.24)],

$$\mathcal{G}_1^{(\beta),*}(w) := \lim_{x \rightarrow -\infty} \left[\sum_{j: a_{\beta,j} > x} b_j(w) \right], \quad \text{where } b_j(w) := \frac{i - w}{(w - a_{\beta,j})(i - a_{\beta,j})}, \quad (4.2)$$

$$\mathcal{G}_1^{(\beta),**, \Im} := - \lim_{x \rightarrow -\infty} \sum_{j: a_{\beta,j} > x} \frac{1}{1 + a_{\beta,j}^2}, \quad (4.3)$$

$$\mathcal{G}_1^{(\beta),**, \Re} := \lim_{x \rightarrow -\infty} \left[\sum_{j: a_{\beta,j} > x} \frac{-a_{\beta,j}}{1 + a_{\beta,j}^2} - \frac{2}{\pi} \sqrt{-x} \right], \quad (4.4)$$

$$\mathcal{G}_2^{(\beta),*}(w) := \lim_{x \rightarrow -\infty} \left[\sum_{j: a_{\beta,j} > x} \eta_j b_j(w) \right], \quad \text{where } \eta_j := \beta^{-1} \zeta_{\beta,j}^2 - 1, \quad (4.5)$$

$$\mathcal{G}_2^{(\beta),**, \Im} := - \lim_{x \rightarrow -\infty} \sum_{j: a_{\beta,j} > x} \frac{\eta_j}{1 + a_{\beta,j}^2}, \quad (4.6)$$

$$\mathcal{G}_2^{(\beta), **, \Re} := \lim_{x \rightarrow -\infty} \sum_{j: a_{\beta, j} > x} \eta_j \frac{-a_{\beta, j}}{1 + a_{\beta, j}^2}. \quad (4.7)$$

We also set

$$\mathcal{G}_\star^{(\beta)}(w) := \mathcal{G}_\star^{(\beta), *} (w) + i\mathcal{G}_\star^{(\beta), **, \Im} + \mathcal{G}_\star^{(\beta), **, \Re}, \quad \star = 1 \text{ or } 2.$$

Then $\mathcal{G}^{(\beta)}(w) = \mathcal{G}_1^{(\beta)}(w) + \mathcal{G}_2^{(\beta)}(w)$.

The $\mathcal{G}_1^{(\beta)}(w)$ part is handled like the $\mathcal{G}_1(w)$ term defined in [19, Equation (8.20)], with [19, Lemma 8.9] replaced by Lemma 4.1 in our case. Indeed, by the estimate of $a_{\beta, j}$ in (4.1), we have that for w in a compact subset of $\mathbb{C} \setminus \{a_{\beta, j} : j \geq 1\}$, uniformly

$$|b_j(w)| = O(j^{-4/3})$$

as $j \rightarrow \infty$, and similarly $(1 + a_{\beta, j}^2)^{-1} = O(j^{-4/3})$ as $j \rightarrow \infty$. Therefore, $\mathcal{G}_1^{(\beta), *} (w)$ is well defined and analytic on $\mathbb{C} \setminus \{a_{\beta, j} : j \geq 1\}$, and the convergence of (4.2) is uniform in a compact subset of $\mathbb{C} \setminus \{a_{\beta, j} : j \geq 1\}$. Similarly, the right-hand side of (4.3) converges absolutely, and the convergence of the right-hand side of (4.4) is obtained by the approximating $a_{\beta, j}$ by $-(3\pi j/2)^{2/3}$ as in [19, Equations (8.21) and (8.22)]. Here we note that in [19, Equation (8.22)], the error of the approximation is controlled as $\mathbf{a}_j + (3\pi j/2)^{2/3} = O(j^\epsilon)$, but it suffices to use the error bound $O(j^{1/6+\epsilon})$ as in (4.1). Hence $\mathcal{G}_1^{(\beta), **, \Im}$ and $\mathcal{G}_1^{(\beta), **, \Re}$ are well defined. We omit the details.

The $\mathcal{G}_2^{(\beta)}(w)$ part is handled like the $\mathcal{G}_2(w)$ term defined in [19, Equation (8.24)]. Condition on a realization of $\{a_{\beta, j}\}$ satisfying (4.1) for all sufficiently large j . By (4.1),

$$\sum_{j=1}^{\infty} \frac{1}{(1 + a_{\beta, j}^2)^2} < \infty, \quad \sum_{j=1}^{\infty} \frac{a_{\beta, j}^2}{(1 + a_{\beta, j}^2)^2} < \infty.$$

Since the random variables η_j are independent, centered, and have finite variance, Kolmogorov's two-series theorem implies that the series on the right-hand sides of (4.6) and (4.7) converge almost surely. Hence $\mathcal{G}_2^{(\beta), **, \Im}$ and $\mathcal{G}_2^{(\beta), **, \Re}$ are well defined almost surely.

It remains to consider $\mathcal{G}_2^{(\beta), *} (w)$. Since $\mathbb{E}|\eta_1| < \infty$ and $(1 + a_{\beta, j}^2)^{-1} = O(j^{-4/3})$, Tonelli's theorem gives, almost surely,

$$\sum_{j=1}^{\infty} \frac{|\eta_j|}{1 + a_{\beta, j}^2} < \infty. \quad (4.8)$$

Let

$$K \Subset \mathbb{C} \setminus \{a_{\beta, j} : j \geq 1\}.$$

By the definition of $b_j(w)$ and (4.1), for all sufficiently large j ,

$$\sup_{w \in K} |b_j(w)| \leq \frac{C_K}{1 + a_{\beta, j}^2}.$$

Consequently, by (4.8), almost surely,

$$\sum_{j=1}^{\infty} |\eta_j| \sup_{w \in K} |b_j(w)| < \infty.$$

The Weierstrass M -test therefore implies that the series on the right-hand side of (4.5) converges absolutely and uniformly on K , almost surely. Since K is arbitrary, $\mathcal{G}_2^{(\beta), *} (w)$ is well defined and analytic on $\mathbb{C} \setminus \{a_{\beta, j} : j \geq 1\}$.

This proves Proposition 1.5.

Remark 4.1. We remark here that based on the convergences proved above, the series

$$\sum_{j=1}^{\infty} \beta^{-1} \xi_{\beta,j}^2 \left(\frac{1}{w - a_{\beta,j}} + \frac{a_{\beta,j}}{1 + a_{\beta,j}^2} \right)$$

converges and is equal to $\mathcal{G}_1^{(\beta),*}(w) + i\mathcal{G}_1^{(\beta),**, \Im} + \mathcal{G}_2^{(\beta),*}(w) + i\mathcal{G}_2^{(\beta),**, \Im}$. This result will be used in the proof of Proposition 2.3.

4.1.3. *Proof of Lemma 1.7 for $\mathcal{G}^{(\beta)}$.*

The proof is similar to the proof of [19, Proposition 8.3]. We write, as in [19, Equation (8.25)],

$$\mathcal{G}^{(\beta)}(w) = \mathcal{G}_1^{***}(w) + \mathcal{G}_2^{(\beta),***}(w) + \mathcal{G}_3^{(\beta),***}(w),$$

where, with η_j defined in (4.5),

$$\begin{aligned} \mathcal{G}_1^{***}(w) &= \lim_{x \rightarrow -\infty} \left[\sum_{j: -(\frac{3\pi j}{2})^{2/3} > x} \frac{1}{w + (\frac{3\pi j}{2})^{2/3}} - \frac{2}{\pi} \sqrt{-x} \right], \\ \mathcal{G}_2^{(\beta),***}(w) &= \sum_{j=1}^{\infty} \frac{\eta_j}{w + (\frac{3\pi j}{2})^{2/3}}, \\ \mathcal{G}_3^{(\beta),***}(w) &= \sum_{j=1}^{\infty} \beta^{-1} \xi_{\beta,j}^2 \frac{(\frac{3\pi j}{2})^{2/3} + a_{\beta,j}}{(w - a_{\beta,j})(w + (\frac{3\pi j}{2})^{2/3})}. \end{aligned}$$

We note that $\mathcal{G}_1^{***}(w)$ is independent of β , and it is the first term on the right-hand side of [19, Equation (8.25)]. [19, Equation (8.29)] yields

$$\mathcal{G}_1^{***}(w) + \sqrt{w} = O(|w|^{-\frac{1}{2}}) \quad \text{as } |w| \rightarrow \infty, \quad \Re w \geq 0.$$

The estimates of $\mathcal{G}_2^{(\beta),***}(w)$ and $\mathcal{G}_3^{(\beta),***}(w)$ are similar to the estimates of the second and third terms on the right-hand side of [19, Equation (8.25)]. We give the key steps of the estimates below and omit some details.

We first show that $\mathcal{G}_3^{(\beta),***}(w) = O(|w|^{-1/4+c})$ for any $c > 0$, in the sense that there exists a random variable $A = A(c)$, such that for all w with $\Re(w) \geq 0$ and $|w| \geq 1$, $|\mathcal{G}_3^{(\beta),***}(w)| \leq A|w|^{-1/4+c}$. To see it, we apply the Borel–Cantelli lemma to $\beta^{-1} \xi_{\beta,j}^2$ and apply Lemma 4.1 to $(3\pi j/2)^{2/3} + a_{\beta,j}$, and find that for every small $\epsilon > 0$, almost surely there exists $C_\epsilon > 0$ such that with $\mathcal{J} = \mathcal{J}(\epsilon)$ defined in Lemma 4.1,

$$\beta^{-1} \xi_{\beta,j}^2 \leq C_\epsilon j^\epsilon, \quad \left| \left(\frac{3\pi j}{2} \right)^{2/3} + a_{\beta,j} \right| \leq j^{\frac{1}{6}+\epsilon}, \quad \text{if } j \geq \mathcal{J}.$$

Then, arguing as in [19, Equation (8.27)] and increasing $\mathcal{J}$ if necessary, we obtain, with $M = \lfloor |w|^{3/2} \rfloor$,

$$\begin{aligned} \left| \mathcal{G}_3^{(\beta),***}(w) \right| &\leq \sum_{j=1}^{\mathcal{J}-1} \beta^{-1} \xi_{\beta,j}^2 \frac{\left| \left(\frac{3\pi j}{2} \right)^{2/3} + a_{\beta,j} \right|}{|w - a_{\beta,j}| \left| w + \left(\frac{3\pi j}{2} \right)^{2/3} \right|} \\ &\quad + C_\epsilon \sum_{j=\mathcal{J}}^M \frac{j^{1/6+2\epsilon}}{\left| w + \left(\frac{3\pi j}{2} \right)^{2/3} \right|^2} + C_\epsilon \sum_{j=M+1}^{\infty} \frac{j^{1/6+2\epsilon}}{\left| w + \left(\frac{3\pi j}{2} \right)^{2/3} \right|^2}. \end{aligned}$$

As $|w| \rightarrow \infty$ with $\Re w \geq 0$, the first term is $O(|w|^{-2})$ almost surely, whereas

$$|w|^{-2} \sum_{j \leq M} j^{1/6+2\epsilon} = O(|w|^{-1/4+3\epsilon})$$

and

$$\sum_{j > M} j^{-7/6+2\epsilon} = O(|w|^{-1/4+3\epsilon}).$$

(We note that, in the setting of [19], [19, Equation (8.17)] yields

$$\left| \left(\frac{3\pi j}{2} \right)^{2/3} + a_j \right| \leq j^\epsilon,$$

without the additional factor $j^{1/6}$ appearing in (4.1). Consequently, the second and third sums in [19, Equation (8.26)] are both bounded by

$$O(|w|^{\frac{3}{2}(2\epsilon - \frac{1}{3})}) = O(|w|^{-1/2+3\epsilon}).$$

Thus, the left-hand side of [19, Equation (8.25)] in fact admits the sharper bound $O(|w|^{-1/2+c})$ for every $c > 0$.)

For $\mathcal{G}_2^{(\beta),***}(w)$, we note that [19] proves with the help of the law of the iterated logarithm that for any independent and identically distributed random variables $\{A_j\}_{j=1}^\infty$ whose mean is 0 and variance finite, there exists a random variable A' such that for any integer M ,

$$\left| \sum_{j=1}^M \frac{A_j}{w + (\frac{3\pi j}{2})^{2/3}} \right| \leq \frac{A'}{|w| + M^{2/3}} \sqrt{M \log \log(M+2)} + \sum_{j=1}^\infty \frac{A' \sqrt{j \log \log(j+2)} j^{-1/3}}{(|w| + j^{2/3})^2},$$

and it tends to 0 as $|w| \rightarrow \infty$, uniformly in M . (In [19], this result is proved with $A_j = \xi_j^2 - 1$ with $\{\xi_j^2\}_{j=1}^\infty$ are independent χ^2 random variables, but the proof depends only on the zero mean and finite variance properties.) Since $\{\eta_j\}$ satisfies the condition for $\{A_j\}$, we find $\mathcal{G}_2^{(\beta),***}(w)$ tends to 0 as $|w| \rightarrow \infty$, uniformly in M .

Thus the proof of Lemma 1.7 for $\mathcal{G}^{(\beta)}$ is complete.

Remark 4.2. The proof of Lemma 1.7 above does not extend [19, Equation (8.3)] to general $\beta > 0$, because our estimate of $\mathcal{G}_3^{(\beta),***}(w)$ is weaker than the desired $o(|w|^{-1/4})$, which is the result that our Lemma 4.1 is weaker than [19, Lemma 8.9] for $\beta = 1$ case.

4.2. Proof of Proposition 2.3. Our proof of Proposition 2.3 follows the same strategy of the proof of [19, Theorem 8.20] that is based on [19, Theorem 8.13].

First, we verify that $\mathcal{G}_N^{(\beta)}(w)$ and $\mathcal{G}^{(\beta)}(w)$ are all in the function space Ω_- defined in [19, Definition 8.10]. To this end, first by Remark 4.1, we use the notation

$$x_j := a_{\beta,j}, \quad w_j := \beta^{-1} \xi_{\beta,j}^2, \quad \gamma := \mathcal{G}_1^{(\beta),**, \Re} + \mathcal{G}_2^{(\beta),**, \Re},$$

where $\{a_{\beta,j}\}_{j=1}^\infty$ are the Airy_β point process defined in (1.12), $\xi_{\beta,j}^2$ are independent χ_β^2 random variables in (2.8), and $\mathcal{G}_1^{(\beta),**, \Re}$ and $\mathcal{G}_2^{(\beta),**, \Re}$ are defined in (4.4) and (4.7) respectively, we have

$$\mathcal{G}^{(\beta)}(w) = \gamma + \sum_{j=1}^\infty w_j \left(\frac{1}{w - x_j} + \frac{x_j}{1 + x_j^2} \right).$$

Hence, $\mathcal{G}^{(\beta)} \in \Omega_-$ almost surely. We note that the condition [19, Equation (8.32)] is due to (4.8). On the other hand, we set

$$x_{j;N} := \begin{cases} N^{2/3}(\mu_j^{(\beta)} - 2), & j \leq N, \\ N^{2/3}(\mu_N^{(\beta)} - 2) - (j - N), & j > N. \end{cases} \quad w_{j;N} := \begin{cases} \beta^{-1}\xi_{\beta,N,j}^2, & j \leq N, \\ 0, & j > N, \end{cases}$$

where $\{\mu_j^{(\beta)}\}_{j=1}^N$ are defined in (1.6) and $\{\xi_{\beta,N,j}^2\}_{j=1}^N$ are independent χ_β^2 random variables in (2.8), and then let

$$\gamma_N := - \sum_{j=1}^N w_{j;N} \frac{x_{j;N}}{1 + x_{j;N}^2} - N^{1/3}.$$

Then

$$\mathcal{G}_N^{(\beta)}(w) = \gamma_N + \sum_{j=1}^{\infty} w_{j;N} \left(\frac{1}{w - x_{j;N}} + \frac{x_{j;N}}{1 + x_{j;N}^2} \right),$$

and it is obvious that $\mathcal{G}_N^{(\beta)} \in \Omega_-$.

Second, we verify the convergence of poles and weights that is required in [19, Equation (8.34)]. By Proposition 1.2, for every fixed M ,

$$(x_{1;N}, \dots, x_{M;N}) = (N^{2/3}(\mu_1^{(\beta)} - 2), \dots, N^{2/3}(\mu_M^{(\beta)} - 2)) \implies (a_{\beta,1}, \dots, a_{\beta,M}).$$

For the weights, we may realize all $\xi_{\beta,N,j}$ and $\xi_{\beta,j}$ on one probability space so that

$$\xi_{\beta,N,j} = \xi_{\beta,j}, \quad 1 \leq j \leq N.$$

Then $w_{j;N} = w_j$ for every fixed j and all large N . Since the weights are independent of the eigenvalues, the above convergences hold jointly in finite-dimensional distributions.

Third, we verify the asymptotics along the imaginary axis for the limit that are required in [19, Equation (8.35)]. Set

$$\phi(R) := -\frac{1+i}{\sqrt{2}}\sqrt{R}.$$

It suffices to prove that

$$\mathcal{G}^{(\beta)}(iR) - \phi(R) \rightarrow 0, \quad \text{almost surely as } R \rightarrow \infty, \quad (4.9)$$

$$\lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E} \left| \mathcal{G}_N^{(\beta)}(iR) - \phi(R) \right|^2 = 0. \quad (4.10)$$

The convergence (4.9) is a direct consequence of Lemma 1.7, and we prove (4.10) below. We recall that the Stieltjes transforms of the empirical spectral measure of $H_N^{(\beta)}$ and the semicircle law (see (1.7)) are

$$m_N(z) := \frac{1}{N} \sum_{j=1}^N \frac{1}{z - \mu_j^{(\beta)}}, \quad m(z) := \int \frac{1}{z - x} d\mu_{\text{SC}}(x) = \frac{z - \sqrt{z^2 - 4}}{2}.$$

Then, as $\eta \in (0, +\infty)$ approaches 0,

$$m(2 + i\eta) = 1 - \frac{1+i}{\sqrt{2}}\sqrt{\eta} + o(\sqrt{\eta}). \quad (4.11)$$

Moreover, by [18, Theorem 1.1 and Remark 1.3], there exist constants $C, \eta_0 > 0$ such that

$$\mathbb{E} [|m_N(2 + i\eta) - m(2 + i\eta)|^2] \leq \frac{C}{N^2\eta^2}, \quad 0 < \eta \leq \eta_0. \quad (4.12)$$

Indeed, analogous to [19, Equation (8.51)],

$$\mathcal{G}_N^{(\beta)}(iR) = N^{1/3} \left(m_N \left(2 + \frac{iR}{N^{2/3}} \right) - 1 \right) + \sum_{j=1}^N \frac{\beta^{-1} \xi_{\beta,N,j}^2 - 1}{iR - x_{j;N}}.$$

Taking the expectation with respect to $\xi_{\beta,N,j}^2$ first, and using that $\{\beta^{-1} \xi_{\beta,N,j}^2 - 1\}_{j=1}^N$ are independent and identically distributed random variables with zero mean and variance $2/\beta$ and are independent of $\{x_{j;N}\}_{j=1}^N$, we obtain, analogous to [19, Equation (8.51)]

$$\mathbb{E} \left| \mathcal{G}_N^{(\beta)}(iR) - \phi(R) \right|^2 \leq 2 \mathbb{E} \left| N^{1/3} \left(m_N \left(2 + \frac{iR}{N^{2/3}} \right) - 1 \right) - \phi(R) \right|^2 + \frac{4}{\beta} \mathbb{E} \sum_{j=1}^N \frac{1}{R^2 + x_{j;N}^2}.$$

Applying (4.12) with $\eta = RN^{-2/3}$, and then using (4.11), we obtain

$$\limsup_{N \rightarrow \infty} \mathbb{E} \left| N^{1/3} \left(m_N \left(2 + \frac{iR}{N^{2/3}} \right) - 1 \right) - \phi(R) \right|^2 \leq \frac{C}{R^2}.$$

Hence the first term vanishes as $R \rightarrow \infty$. For the second term, the resolvent identity gives

$$\sum_{j=1}^N \frac{1}{R^2 + x_{j;N}^2} = \frac{N^{1/3}}{2iR} \left[m_N \left(2 - \frac{iR}{N^{2/3}} \right) - m_N \left(2 + \frac{iR}{N^{2/3}} \right) \right] = -\frac{N^{1/3}}{R} \Im m_N \left(2 + \frac{iR}{N^{2/3}} \right),$$

Applying (4.12) again with $\eta = RN^{-2/3}$, and using (4.11), we obtain

$$\limsup_{N \rightarrow \infty} \mathbb{E} \sum_{j=1}^N \frac{1}{R^2 + x_{j;N}^2} \leq \frac{1}{\sqrt{2}R} + \frac{C}{R^2}.$$

which tends to zero as $R \rightarrow \infty$.

We have thus verified all the hypotheses of [19, Theorem 8.13] for $\mathcal{G}_N^{(\beta)}$ and $\mathcal{G}^{(\beta)}$, and complete the proof of Proposition 2.3.

4.3. Proof of Proposition 1.6 and Lemma 1.7 for $\mathcal{G}_\Theta^{(1)}$ and $\mathcal{G}_\Theta^{(2)}$. In this subsection, we prove Proposition 1.6 and the part of Lemma 1.7 concerning $\mathcal{G}_\Theta^{(1)}$ and $\mathcal{G}_\Theta^{(2)}$. The proofs are similar to those of [19, Theorem 8.1 and Proposition 8.3]; it suffices to replace the Airy points there by the multivariate stochastic Airy operator edge points $a_{\beta,j}^{(\Theta)}$. This follows from the rigidity estimate below.

We prove the case $\beta = 1$; the case $\beta = 2$ is parallel. For notational simplicity, we write

$$a_j := a_{1,j}, \quad a_j^\Theta := a_{1,j}^\Theta, \quad \mathcal{G}_\Theta := \mathcal{G}_\Theta^{(1)}.$$

When $l = 0$, we use the convention $a_j^\Theta = a_j^{(\cdot)} = a_j$.

Lemma 4.2. *Let $l \geq 0$ and $\Theta = (\theta_1, \dots, \theta_l)$. For every $\varepsilon > 0$, there exists a random variable $\mathcal{J}' = \mathcal{J}'(\varepsilon)$ such that almost surely,*

$$\left| a_j^\Theta + \left(\frac{3\pi j}{2} \right)^{2/3} \right| \leq j^\varepsilon, \quad j > \mathcal{J}'. \quad (4.13)$$

Proof. If $l = 0$, this is [19, Lemma 8.9]. We assume $l > 0$.

We use finite- N interlacing together with joint convergence. Let H_N be the GOE matrix, let $\mu_1 \geq \dots \geq \mu_N$ be its eigenvalues, and let $\lambda_1 \geq \dots \geq \lambda_N$ be the eigenvalues of

$$H_N + \text{diag}(d_1, \dots, d_l, 0, \dots, 0), \quad d_i = 1 - \theta_i N^{-1/3}.$$

Since $d_i > 0$ for all sufficiently large N , the rank- l interlacing inequalities give

$$\mu_j \leq \lambda_j \leq \mu_{j-l}, \quad j > l,$$

with the convention that $\mu_m = +\infty$ for $m \leq 0$. Equivalently, at the soft edge,

$$N^{2/3}(\mu_j - 2) \leq N^{2/3}(\lambda_j - 2) \leq N^{2/3}(\mu_{j-l} - 2), \quad j > l.$$

By the soft-edge convergence of the GOE and by Proposition 1.4, for each fixed M the two marginal vectors

$$(N^{2/3}(\mu_1 - 2), \dots, N^{2/3}(\mu_M - 2)), \quad (N^{2/3}(\lambda_1 - 2), \dots, N^{2/3}(\lambda_M - 2))$$

are tight and converge respectively to

$$(a_1, \dots, a_M), \quad (a_1^\Theta, \dots, a_M^\Theta).$$

Hence, after passing to a joint subsequential limit, we obtain a coupling of the two limiting point processes. Since the finite- N interlacing inequalities define a closed condition, they pass to the limit, yielding

$$a_j \leq a_j^\Theta \leq a_{j-l}, \quad l < j \leq M.$$

A diagonal argument in M gives this coupling for all $j > l$.

Applying [19, Lemma 8.9] with $\varepsilon/2$, we have almost surely, for all large j ,

$$\left| a_j + \left(\frac{3\pi j}{2} \right)^{2/3} \right| + \left| a_{j-l} + \left(\frac{3\pi(j-l)}{2} \right)^{2/3} \right| \leq 2j^{\varepsilon/2}.$$

Since l is fixed,

$$\left(\frac{3\pi(j-l)}{2} \right)^{2/3} = \left(\frac{3\pi j}{2} \right)^{2/3} + O(j^{-1/3}).$$

Together with

$$a_j \leq a_j^\Theta \leq a_{j-l},$$

this gives

$$\left| a_j^\Theta + \left(\frac{3\pi j}{2} \right)^{2/3} \right| \leq j^\varepsilon$$

for all sufficiently large j , after increasing the random threshold. $\square$

Proof of Proposition 1.6. The proof is the same as the proof of [19, Theorem 8.1]. Indeed, in that proof the only input on the pole locations is the rigidity estimate for the Airy points, and the only input on the weights is that, after subtracting one, they are independent centered random variables with finite variance. By Lemma 4.2, the same rigidity estimate holds for a_j^Θ . Therefore the argument in [19, Theorem 8.1] applies verbatim after replacing a_j by a_j^Θ . This proves the existence of the limit in (1.14), uniformly on compact subsets avoiding the poles. $\square$

Proof of Lemma 1.7 for $\mathcal{G}_\Theta^{(\beta)}$, $\beta \in \{1, 2\}$. We follow the proof of [19, Proposition 8.3]. By Lemma 4.2 for $\beta = 1$, and by its parallel GUE version for $\beta = 2$, for every $\varepsilon > 0$, almost surely,

$$a_{\beta,j}^\Theta = -\left(\frac{3\pi j}{2}\right)^{2/3} + O(j^\varepsilon), \quad j \rightarrow \infty.$$

Moreover, the exponential tail of the chi-square distribution and the Borel–Cantelli lemma give

$$\beta^{-1}\xi_{\beta,j}^2 = O(j^\varepsilon) \quad \text{almost surely.}$$

Using

$$\frac{\beta^{-1}\xi_{\beta,j}^2}{w - a_{\beta,j}^\Theta} = \frac{1}{w + \left(\frac{3\pi j}{2}\right)^{2/3}} + \frac{\beta^{-1}\xi_{\beta,j}^2 - 1}{w + \left(\frac{3\pi j}{2}\right)^{2/3}} + \beta^{-1}\xi_{\beta,j}^2 \frac{\left(\frac{3\pi j}{2}\right)^{2/3} + a_{\beta,j}^\Theta}{(w - a_{\beta,j}^\Theta) \left(w + \left(\frac{3\pi j}{2}\right)^{2/3}\right)},$$

we obtain, as in [19, Equation (8.25)], the decomposition

$$\begin{aligned} \mathcal{G}_\Theta^{(\beta)}(w) &= \lim_{y \rightarrow -\infty} \left[\sum_{\left(\frac{3\pi j}{2}\right)^{2/3} < -y} \frac{1}{w + \left(\frac{3\pi j}{2}\right)^{2/3}} - \frac{2}{\pi} \sqrt{-y} \right] \\ &\quad + \sum_{j=1}^{\infty} \frac{\beta^{-1}\xi_{\beta,j}^2 - 1}{w + \left(\frac{3\pi j}{2}\right)^{2/3}} \\ &\quad + \sum_{j=1}^{\infty} \beta^{-1}\xi_{\beta,j}^2 \frac{\left(\frac{3\pi j}{2}\right)^{2/3} + a_{\beta,j}^\Theta}{(w - a_{\beta,j}^\Theta) \left(w + \left(\frac{3\pi j}{2}\right)^{2/3}\right)}. \end{aligned} \quad (4.14)$$

Here the rigidity estimate above plays the same role as [19, Lemma 8.9] in replacing the cutoff $a_{\beta,j}^\Theta > y$ by $\left(\frac{3\pi j}{2}\right)^{2/3} < -y$.

As in [19, Equations (8.28)–(8.29)], comparison with the corresponding integral gives, uniformly for $\Re w \geq 0$,

$$\lim_{y \rightarrow -\infty} \left[\sum_{\left(\frac{3\pi j}{2}\right)^{2/3} < -y} \frac{1}{w + \left(\frac{3\pi j}{2}\right)^{2/3}} - \frac{2}{\pi} \sqrt{-y} \right] = -\sqrt{w} + O(|w|^{-1/2}).$$

Next, the law of the iterated logarithm gives

$$\sum_{k=1}^j (\beta^{-1}\xi_{\beta,k}^2 - 1) = O(\sqrt{j \log \log(j+2)}) \quad \text{almost surely.}$$

The second term in (4.14) is treated by the summation-by-parts argument used for the second sum in [19, Equation (8.25), in the paragraph between Equations (8.27) and (8.28)]. It follows that

$$\sum_{j=1}^{\infty} \frac{\beta^{-1}\xi_{\beta,j}^2 - 1}{w + \left(\frac{3\pi j}{2}\right)^{2/3}} \rightarrow 0$$

almost surely, uniformly as $|w| \rightarrow \infty$ with $\Re w \geq 0$.

For the last term in (4.14), we use the estimate [19, Equation (8.26)] and the three-range decomposition in [19, Equation (8.27)], splitting the sum at $j = \lfloor |w|^{3/2} \rfloor$. For $0 < \varepsilon < 1/12$, the rigidity estimate and the chi-square tail bound imply

$$\left| \sum_{j=1}^{\infty} \beta^{-1} \xi_{\beta,j}^2 \frac{\left(\frac{3\pi j}{2}\right)^{2/3} + a_{\beta,j}^{\Theta}}{\left(w - a_{\beta,j}^{\Theta}\right) \left(w + \left(\frac{3\pi j}{2}\right)^{2/3}\right)} \right| \leq O(|w|^{-2}) + C|w|^{-2} \sum_{j \leq |w|^{3/2}} j^{2\varepsilon} + C \sum_{j > |w|^{3/2}} j^{-4/3+2\varepsilon} \\ = O(|w|^{-1/2+3\varepsilon}) = o(|w|^{-1/4}).$$

The same calculation after m differentiations gives

$$\frac{d^m}{dw^m} \sum_{j=1}^{\infty} \beta^{-1} \xi_{\beta,j}^2 \frac{\left(\frac{3\pi j}{2}\right)^{2/3} + a_{\beta,j}^{\Theta}}{\left(w - a_{\beta,j}^{\Theta}\right) \left(w + \left(\frac{3\pi j}{2}\right)^{2/3}\right)} = O(|w|^{-m-1/2+3\varepsilon}), \quad m = 1, 2.$$

We now prove the joint fluctuation statement directly from (4.14). Termwise differentiation is justified by local uniform convergence away from the poles. The estimates above and their differentiated versions give

$$\begin{pmatrix} x^{1/4}(\mathcal{G}_{\Theta}^{(\beta)}(x) + \sqrt{x}) \\ x^{5/4} \left((\mathcal{G}_{\Theta}^{(\beta)})'(x) + \frac{1}{2\sqrt{x}} \right) \end{pmatrix} = \sum_{j=1}^{\infty} (\beta^{-1} \xi_{\beta,j}^2 - 1) \begin{pmatrix} \frac{x^{1/4}}{x + \left(\frac{3\pi j}{2}\right)^{2/3}} \\ -\frac{x^{5/4}}{\left(x + \left(\frac{3\pi j}{2}\right)^{2/3}\right)^2} \end{pmatrix} + o_{\mathbb{P}}(1).$$

For every $p > 3/2$, the same sum-integral comparison as in [19, Equation (8.30)] gives

$$x^{p-3/2} \sum_{j=1}^{\infty} \frac{1}{\left(x + \left(\frac{3\pi j}{2}\right)^{2/3}\right)^p} \rightarrow \frac{1}{\pi} \int_0^{\infty} \frac{u^{1/2}}{(1+u)^p} du.$$

The case $p = 2$ is precisely the variance calculation in [19, Equation (8.30)]; the cases $p = 3, 4$ give the covariance and the variance of the derivative component. Since

$$\text{Var}(\beta^{-1} \xi_{\beta,j}^2 - 1) = \frac{2}{\beta},$$

the covariance matrix of the centered sum converges to

$$\frac{1}{\beta} \begin{pmatrix} 1 & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{8} \end{pmatrix}.$$

The largest coefficient in the triangular array is $O(x^{-3/4})$, while

$$\mathbb{E} |\beta^{-1} \xi_{\beta,j}^2 - 1|^3 < \infty.$$

Thus the Lyapunov condition and the Cramér–Wold device give

$$\begin{pmatrix} x^{1/4}(\mathcal{G}_{\Theta}^{(\beta)}(x) + \sqrt{x}) \\ x^{5/4} \left((\mathcal{G}_{\Theta}^{(\beta)})'(x) + \frac{1}{2\sqrt{x}} \right) \end{pmatrix} \Rightarrow \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \frac{1}{\beta} \begin{pmatrix} 1 & -\frac{1}{4} \\ -\frac{1}{4} & \frac{1}{8} \end{pmatrix} \right).$$

Since the second component is tight,

$$x^{1/4} \left[-\frac{1}{(\mathcal{G}_\Theta^{(\beta)})'(x)} - 2\sqrt{x} \right] = 4x^{5/4} \left((\mathcal{G}_\Theta^{(\beta)})'(x) + \frac{1}{2\sqrt{x}} \right) + o_{\mathbb{P}}(1).$$

It follows that

$$\left(\begin{array}{c} x^{1/4}(\mathcal{G}_\Theta^{(\beta)}(x) + \sqrt{x}) \\ x^{1/4} \left[-\frac{1}{(\mathcal{G}_\Theta^{(\beta)})'(x)} - 2\sqrt{x} \right] \end{array} \right) \Rightarrow \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \frac{1}{\beta} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \right).$$

Finally, the three estimates following (4.14) imply that, almost surely,

$$\mathcal{G}_\Theta^{(\beta)}(w) + \sqrt{w} \rightarrow 0$$

uniformly as $|w| \rightarrow \infty$, $\Re w \geq 0$. Applying Cauchy's formula to $\mathcal{G}_\Theta^{(\beta)}(w) + \sqrt{w}$ on $|w - x| = x/2$ gives

$$(\mathcal{G}_\Theta^{(\beta)})'(x) = -\frac{1}{2\sqrt{x}} + o(x^{-1}), \quad (\mathcal{G}_\Theta^{(\beta)})''(x) = \frac{1}{4x^{3/2}} + o(x^{-2}),$$

which completes the proof. $\square$

4.4. Proof of Proposition 2.5. The proof is based on [19, Theorem 8.13], the same as the proof of Proposition 2.3 and the proof of [19, Theorem 8.20]. We need the following lemma whose proof will be given in Section 5.

Lemma 4.3 (Spherical second and fourth moments in a subspace). *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and put*

$$\beta_{\mathbb{F}} := \dim_{\mathbb{R}} \mathbb{F} = \begin{cases} 1, & \mathbb{F} = \mathbb{R}, \\ 2, & \mathbb{F} = \mathbb{C}. \end{cases}$$

Equip $\mathbb{F}^N$ with the standard Euclidean/Hermitian inner product. Let $V \subset \mathbb{F}^N$ be an m -dimensional $\mathbb{F}$ -linear subspace, and let P be the matrix representing the orthogonal projection onto V . If v is a random unit vector in V uniformly distributed on the unit sphere of V , then, with v^ understood as $v^\top$ if $\mathbb{F} = \mathbb{R}$,*

$$\mathbb{E}[vv^*] = \frac{1}{m}P. \quad (4.15)$$

Moreover, for any $N \times N$ matrices A and B that are real symmetric if $\mathbb{F} = \mathbb{R}$ and are Hermitian if $\mathbb{F} = \mathbb{C}$,

$$\mathbb{E}[(v^*Av)(v^*Bv)] = \frac{\text{tr}(PAP)\text{tr}(PBP) + \frac{2}{\beta_{\mathbb{F}}}\text{tr}(PAPBP)}{m(m + \frac{2}{\beta_{\mathbb{F}}})}. \quad (4.16)$$

Proof of Proposition 2.5. For notational simplicity, we adapt the short-handed notation as in (2.12) in Section 2.3. We also make use of notation in Section 4.3.

First, we verify that $\tilde{\mathcal{G}}_{N,r}(w)$ and $\mathcal{G}_{(r-1),r^c}(w)$ are all in the function space Ω_- defined in [19, Definition 8.10]. To this end, we denote

$$x_j := a_j^{\Theta^{(r)}}, \quad w_j := \xi_j^2, \quad \gamma = \mathcal{G}_{(r-1),r^c,1}^{**, \Re} + \mathcal{G}_{(r-1),r^c,2}^{**, \Re},$$

where $(\xi_j)_{j \geq 1}$ are i.i.d. standard Gaussian random variables, independent of the limiting edge eigenvalue process, and

$$\mathcal{G}_{(r-1),r^c,1}^{**, \Re} = \lim_{x \rightarrow -\infty} \left[\sum_{j: x_j > x} \frac{-x_j}{1+x_j^2} - \frac{2}{\pi} \sqrt{-x} \right], \quad \mathcal{G}_{(r-1),r^c,2}^{**, \Re} = \lim_{x \rightarrow -\infty} \sum_{j: x_j > x} (\xi_j^2 - 1) \frac{-x_j}{1+x_j^2}.$$

Since $a_j^{\Theta(r)}$ satisfies inequality (4.13) by Lemma 4.2, which is analogous to the inequality (4.1) satisfied by $a_{\beta,j}$, by arguments in the proof of Proposition 1.5, especially the well-definedness of $\mathcal{G}_1^{(\beta), **, \Re}$ and $\mathcal{G}_2^{(\beta), **, \Re}$, we have that $\mathcal{G}_{(r-1),r^c,1}^{**, \Re}$ and $\mathcal{G}_{(r-1),r^c,2}^{**, \Re}$ are well-defined random variables. As in the corresponding Ω_- representation for $\mathcal{G}^{(\beta)}$, we have

$$\mathcal{G}_{(r-1),r^c}(w) = \gamma + \sum_{j=1}^{\infty} w_j \left(\frac{1}{w - x_j} + \frac{x_j}{1+x_j^2} \right).$$

Thus $\mathcal{G}_{(r-1),r^c} \in \Omega_-$ almost surely. On the other hand, we set

$$x_{j;N} := \begin{cases} a_{j,N}, & j \leq N, \\ a_{N,N} - (j - N), & j > N. \end{cases} \quad w_{j;N} := \begin{cases} N |\langle e_r, \hat{\mathbf{u}}_{j,N} \rangle|^2, & j \leq N, \\ 0, & j > N, \end{cases}$$

where $\{a_{j,N}\}_{j=1}^N$ and $\{N |\langle e_r, \hat{\mathbf{u}}_{j,N} \rangle|^2\}_{j=1}^N$ are defined in (2.13), and then let

$$\gamma_N := - \sum_{j=1}^N w_{j;N} \frac{x_{j;N}}{1+x_{j;N}^2} - N^{1/3}.$$

Then

$$\tilde{\mathcal{G}}_{N,r}(w) = \gamma_N + \sum_{j=1}^{\infty} w_{j;N} \left(\frac{1}{w - x_{j;N}} + \frac{x_{j;N}}{1+x_{j;N}^2} \right),$$

and it is obvious that $\tilde{\mathcal{G}}_{N,r} \in \Omega_-$.

Second, we verify the convergence of poles and weights that is required in [19, Equation (8.34)]. This follows from Lemma 2.4.

Set

$$\phi(R) := -\frac{1+i}{\sqrt{2}} \sqrt{R}. \quad (4.17)$$

Third, we verify the asymptotics along the imaginary axis for the limit that are required in [19, Equation (8.35)]. With $\phi(R)$ defined in (4.17), by Lemma 1.7

$$\mathcal{G}_{(r-1),r^c}(iR) - \phi(R) \rightarrow 0 \quad \text{almost surely as } R \rightarrow \infty.$$

It remains to show that

$$\lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E} \left| \tilde{\mathcal{G}}_{N,r}(iR) - \phi(R) \right|^2 = 0. \quad (4.18)$$

For the GOE matrix H_N , set

$$m_N(z) := \frac{1}{N} \sum_{j=1}^N \frac{1}{z - \mu_j}, \quad m(z) := G_{\text{SC}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}.$$

As $\eta \downarrow 0$,

$$m(2 + i\eta) = 1 - \frac{1+i}{\sqrt{2}} \sqrt{\eta} + o(\sqrt{\eta}). \quad (4.19)$$

Moreover, by the local L^2 estimate for the Gaussian β -ensemble [18], for every $\varepsilon > 0$, there is a constant $C = C(\varepsilon) > 0$ such that

$$\mathbb{E}[|m_N(2 + i\eta) - m(2 + i\eta)|^2] \leq \frac{C}{N^2\eta^2}, \quad N^{-2/3} < \eta < N^{-2/3+\varepsilon}. \quad (4.20)$$

We define

$$\widehat{m}_N(z) = \frac{1}{N} \sum_{j=1}^N \frac{1}{z - \widehat{\mu}_{j,N}}. \quad (4.21)$$

where $\{\widehat{\mu}_{j,N} : 1 \leq j \leq N\}$ are the eigenvalues of

$$\widehat{H}_N := H_N + \sum_{j=1}^{r-1} d_j e_j e_j^\top. \quad (4.22)$$

By the classical interlacing property for eigenvalues under a rank-1 (or finite-rank) perturbation, we have

$$\widehat{\mu}_{j,N} \geq \mu_j \geq \widehat{\mu}_{j+r-1,N}, \quad 1 \leq j \leq N - r + 1,$$

where $\{\mu_j\}_{j=1}^N$ are the eigenvalues of H_N . By the rank inequality for Stieltjes transforms, for every $\eta > 0$,

$$|\widehat{m}_N(2 + i\eta) - m_N(2 + i\eta)| \leq \frac{C_r}{N\eta}.$$

Combining this bound with (4.20), we obtain

$$\mathbb{E}[|\widehat{m}_N(2 + i\eta) - m(2 + i\eta)|^2] \leq \frac{C_r}{N^2\eta^2}, \quad N^{-2/3} < \eta < N^{-2/3+\varepsilon}. \quad (4.23)$$

We define the vectors

$$\alpha_{j,N} := (\langle e_j, \widehat{\mathbf{u}}_{1,N} \rangle, \dots, \langle e_j, \widehat{\mathbf{u}}_{N,N} \rangle)^\top \in \mathbb{R}^N, \quad j = 1, \dots, N,$$

and set

$$\mathcal{F}_{r-1,N} := \sigma(\widehat{\mu}_{1,N}, \dots, \widehat{\mu}_{N,N}, \alpha_{1,N}, \dots, \alpha_{r-1,N}),$$

where $\widehat{\mu}_{1,N}, \dots, \widehat{\mu}_{N,N}$ are the eigenvalues of $\widehat{H}_N$ defined in (4.22), and $\alpha_{j,N}$ is defined above. Since the vectors $\alpha_{j,N}$ are orthonormal, the operator P_N defined by the matrix

$$P_{r-1,N} := I - \sum_{j=1}^{r-1} \alpha_{j,N} \alpha_{j,N}^\top$$

is the orthogonal projection onto

$$\mathcal{V}_{r-1,N} := \left(\text{Span}\{\alpha_{1,N}, \dots, \alpha_{r-1,N}\} \right)^\perp.$$

By the residual orthogonal-invariance argument in the proof of Lemma 2.4, conditionally on $\mathcal{F}_{r-1,N}$, the vector $\alpha_{r,N}$, modulo the irrelevant signs of the eigenvectors, is uniformly distributed on the unit sphere of $\mathcal{V}_{r-1,N}$, whose dimension is $N - r + 1$.

Fix $R > 0$, and define

$$d_{j,N}(R) := iR - a_{j,N}, \quad D_N(R) := \text{diag}\left(\frac{1}{d_{1,N}(R)}, \dots, \frac{1}{d_{N,N}(R)}\right).$$

where $a_{j,N} := N^{2/3}(\widehat{\mu}_{j,N} - 2)$, as in (2.13). Then

$$\widetilde{\mathcal{G}}_{N,r}(iR) + N^{1/3} = N \alpha_{r,N}^\top D_N(R) \alpha_{r,N}. \quad (4.24)$$

Since $D_N(R)$ is diagonal, the signs of the eigenvectors do not affect the quadratic form below. Hence, by (4.15) in Lemma 4.3, conditionally on $\mathcal{F}_{r-1,N}$,

$$\mathbb{E}[\alpha_{r,N}^\top D_N(R) \alpha_{r,N} \mid \mathcal{F}_{r-1,N}] = \frac{1}{N-r+1} \operatorname{tr}(P_{r-1,N} D_N(R)).$$

Therefore, (4.24) implies

$$\mathbb{E}[\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}] = \frac{N}{N-r+1} \operatorname{tr}(P_{r-1,N} D_N(R)) - N^{1/3}.$$

We denote

$$A_N(R) := \operatorname{tr} D_N(R) - N^{1/3}.$$

Since

$$\operatorname{tr}(P_{r-1,N} D_N(R)) = \operatorname{tr} D_N(R) - \operatorname{tr} \sum_{j=1}^{r-1} \alpha_{j,N}^\top D_N(R) \alpha_{j,N},$$

and $|d_{m,N}(R)| \geq R$, we get

$$|E[\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}] - A_N(R)| \leq \frac{r}{N} |\operatorname{tr} D_N(R)| + \frac{r}{R}$$

for all large N . On the other hand,

$$A_N(R) = N^{1/3} \hat{m}_N \left(2 + \frac{iR}{N^{2/3}} \right) - N^{1/3},$$

where $\hat{m}_N$ is defined in (4.21). Using the estimate (4.23) and (4.19), and arguing as in the proof of [19, Theorem 8.20], we obtain

$$\lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E} \left| E[\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}] - \phi(R) \right|^2 = 0. \quad (4.25)$$

Since we have

$$\mathbb{E} \left| \tilde{\mathcal{G}}_{N,r}(iR) - \phi(R) \right|^2 = \mathbb{E} |E[\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}] - \phi(R)|^2 + \mathbb{E} \operatorname{Var}(\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}). \quad (4.26)$$

It remains to estimate the conditional variance $\operatorname{Var}(\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N})$. Set

$$X_N(R) := N \alpha_{r,N}^\top D_N(R) \alpha_{r,N}, \quad n_N := N - r + 1.$$

Since

$$\tilde{\mathcal{G}}_{N,r}(iR) = X_N(R) - N^{1/3},$$

we have

$$\operatorname{Var}(\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}) = \operatorname{Var}(X_N(R) \mid \mathcal{F}_{r-1,N}).$$

Write

$$D_N(R) = D_N^{\Re}(R) + i D_N^{\Im}(R),$$

where $D_N^{\Re}(R) = \Re D_N(R)$ and $D_N^{\Im}(R) = \Im D_N(R)$ are real diagonal, hence self-adjoint, matrices. Applying Lemma 4.3 separately to $D_N^{\Re}(R)$ and $D_N^{\Im}(R)$, we obtain

$$\operatorname{Var}(X_N(R) \mid \mathcal{F}_{r-1,N}) \leq C_r (\|P_{r-1,N} D_N^{\Re}(R) P_{r-1,N}\|_{HS}^2 + \|P_{r-1,N} D_N^{\Im}(R) P_{r-1,N}\|_{HS}^2).$$

Since

$$\|P_{r-1,N} D_N^{\Re}(R) P_{r-1,N}\|_{HS}^2 + \|P_{r-1,N} D_N^{\Im}(R) P_{r-1,N}\|_{HS}^2 \leq \|D_N(R)\|_{HS}^2,$$

we get

$$\text{Var}(X_N(R) \mid \mathcal{F}_{r-1,N}) \leq C_r \sum_{m=1}^N \frac{1}{|iR - a_{m,N}|^2}.$$

Here $\|A\|_{HS} := (\text{tr}(AA^*))^{1/2}$. Hence

$$\text{Var}(\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}) \leq C_r \sum_{m=1}^N \frac{1}{|iR - a_{m,N}|^2}.$$

Using

$$\sum_{m=1}^N \frac{1}{|iR - a_{m,N}|^2} = \frac{N^{1/3}}{2iR} \left[\hat{m}_N \left(2 - \frac{iR}{N^{2/3}} \right) - \hat{m}_N \left(2 + \frac{iR}{N^{2/3}} \right) \right],$$

and the same local L^2 estimate again, we conclude that

$$\lim_{R \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathbb{E} \text{Var}(\tilde{\mathcal{G}}_{N,r}(iR) \mid \mathcal{F}_{r-1,N}) = 0,$$

and together with (4.25) and (4.26), we verify (4.18).

Thus all the three conditions required by [19, Theorem 8.13] hold, and [19, Theorem 8.13] yields Proposition 2.5. $\square$

5. PROOF OF TWO LINEAR ALGEBRAIC RESULTS

Here we prove Lemma 1.11 and a technical linear algebraic result used in the proof of Theorems 1.8 and 1.9.

Proof of Lemma 1.11. For notational simplicity, we only prove the lemma in the real symmetric setting. The Hermitian setting is parallel.

Recall that $\mathbf{x}_k$ is the normalized eigenvector of $Y = X + d\mathbf{w}\mathbf{w}^\top$ associated with λ_k . Since

$$(X + d\mathbf{w}\mathbf{w}^\top)\mathbf{x}_k = \lambda_k \mathbf{x}_k,$$

we have

$$(X - \lambda_k I)\mathbf{x}_k = -d(\mathbf{w}^\top \mathbf{x}_k)\mathbf{w}.$$

Because $\lambda_k \notin \{\mu_j\}_{j=1}^N = \sigma(X)$, the matrix $X - \lambda_k I$ is invertible, and hence $\mathbf{x}_k$ is proportional to $(X - \lambda_k I)^{-1}\mathbf{w}$. More precisely, up to a sign,

$$\mathbf{x}_k = \frac{(X - \lambda_k I)^{-1}\mathbf{w}}{\sqrt{\mathbf{w}^\top (X - \lambda_k I)^{-2}\mathbf{w}}}.$$

Since λ_k is an eigenvalue of Y , the matrix determinant lemma [30, Theorem 18.1.1] gives

$$\begin{aligned} 0 &= \det(\lambda_k I - Y) = \det(\lambda_k I - X) \det(I - d(\lambda_k I - X)^{-1}\mathbf{w}\mathbf{w}^\top) \\ &= \det(\lambda_k I - X) \left(1 - d\mathbf{w}^\top (\lambda_k I - X)^{-1}\mathbf{w} \right). \end{aligned}$$

Since $\det(\lambda_k I - X) \neq 0$, it follows that (1.27) holds. Hence, (1.28) is also proved. $\square$

Proof of Lemma 4.3. By orthogonal/unitary invariance, we may assume

$$V = \text{Span}_{\mathbb{R}}\{e_1, \dots, e_m\}, \quad P = I_m \oplus 0_{N-m}.$$

Let $\zeta_1, \dots, \zeta_m$ be i.i.d. standard $\mathbb{F}$ -Gaussian variables, real standard normal if $\mathbb{F} = \mathbb{R}$, and standard complex normal if $\mathbb{F} = \mathbb{C}$. Then

$$v = \frac{(\zeta_1, \dots, \zeta_m, 0, \dots, 0)^\top}{(\sum_{i=1}^m |\zeta_i|^2)^{1/2}}$$

is uniformly distributed on the unit sphere of V . This gives that for $1 \leq a, b \leq m$,

$$\mathbb{E}[v_a v_b] = \frac{\delta_{ab}}{m}, \quad (\mathbb{F} = \mathbb{R}), \quad \mathbb{E}[\bar{v}_a v_b] = \frac{\delta_{ab}}{m}, \quad (\mathbb{F} = \mathbb{C}),$$

and hence (4.15).

For the fourth moment, the standard spherical moment identities are

$$\begin{aligned} \mathbb{E}[v_a v_b v_c v_d] &= \frac{\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}}{m(m+2)} & (\mathbb{F} = \mathbb{R}), \\ \mathbb{E}[\bar{v}_a v_b \bar{v}_c v_d] &= \frac{\delta_{ab}\delta_{cd} + \delta_{ad}\delta_{bc}}{m(m+1)} & (\mathbb{F} = \mathbb{C}). \end{aligned}$$

Contracting these identities with the entries of the compressions PAP and PBP , and using that A, B are self-adjoint, gives

$$\mathbb{E}[(v^* A v)(v^* B v)] = \begin{cases} \frac{\text{tr}(PAP) \text{tr}(PBP) + 2 \text{tr}(PAPBP)}{m(m+2)}, & \mathbb{F} = \mathbb{R}, \\ \frac{\text{tr}(PAP) \text{tr}(PBP) + \text{tr}(PAPBP)}{m(m+1)}, & \mathbb{F} = \mathbb{C}. \end{cases}$$

This is exactly (4.16). □

REFERENCES

- [1] G. W. Anderson, A. Guionnet, and O. Zeitouni. *An introduction to random matrices*, volume 118 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2010.
- [2] L. Ashbury-Bridgwood. *Random Canonical Products and the Secular Function of the Stochastic Airy Operator*. ProQuest LLC, Ann Arbor, MI, 2022. Thesis (Ph.D.)—University of Toronto (Canada).
- [3] Z. Bai and J.-f. Yao. Central limit theorems for eigenvalues in a spiked population model. *Ann. Inst. Henri Poincaré Probab. Stat.*, 44(3):447–474, 2008.
- [4] J. Baik, G. Ben Arous, and S. P\'ech\'e. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Ann. Probab.*, 33(5):1643–1697, 2005.
- [5] J. Baik and J. W. Silverstein. Eigenvalues of large sample covariance matrices of spiked population models. *J. Multivariate Anal.*, 97(6):1382–1408, 2006.
- [6] J. Baik and D. Wang. On the largest eigenvalue of a Hermitian random matrix model with spiked external source I. Rank 1 case. *Int. Math. Res. Not. IMRN*, (22):5164–5240, 2011.
- [7] J. Baik and D. Wang. On the largest eigenvalue of a Hermitian random matrix model with spiked external source II: Higher rank cases. *Int. Math. Res. Not. IMRN*, (14):3304–3370, 2013.
- [8] Z. Bao, X. Ding, J. Wang, and K. Wang. Statistical inference for principal components of spiked covariance matrices. *Ann. Statist.*, 50(2):1144–1169, 2022.
- [9] Z. Bao, X. Ding, and K. Wang. Singular vector and singular subspace distribution for the matrix denoising model. *Ann. Statist.*, 49(1):370–392, 2021.
- [10] Z. Bao and D. Wang. Eigenvector distribution in the critical regime of BBP transition. *Probab. Theory Related Fields*, 182(1-2):399–479, 2022.
- [11] F. Benaych-Georges, A. Guionnet, and M. Maida. Fluctuations of the extreme eigenvalues of finite rank deformations of random matrices. *Electron. J. Probab.*, 16:no. 60, 1621–1662, 2011.
- [12] F. Benaych-Georges and R. R. Nadakuditi. The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices. *Adv. Math.*, 227(1):494–521, 2011.

- [13] F. Benaych-Georges and R. R. Nadakuditi. The singular values and vectors of low rank perturbations of large rectangular random matrices. *J. Multivariate Anal.*, 111:120–135, 2012.
- [14] A. Bloemendal, A. Knowles, H.-T. Yau, and J. Yin. On the principal components of sample covariance matrices. *Probab. Theory Related Fields*, 164(1-2):459–552, 2016.
- [15] A. Bloemendal and B. Virág. Limits of spiked random matrices I. *Probab. Theory Related Fields*, 156(3-4):795–825, 2013.
- [16] A. Bloemendal and B. Virág. Limits of spiked random matrices II. *Ann. Probab.*, 44(4):2726–2769, 2016.
- [17] P. Bourgade, L. Erdős, and H.-T. Yau. Edge universality of beta ensembles. *Comm. Math. Phys.*, 332(1):261–353, 2014.
- [18] P. Bourgade, K. Mody, and M. Pain. Optimal local law and central limit theorem for β -ensembles. *Comm. Math. Phys.*, 390(3):1017–1079, 2022.
- [19] A. Bykhovskaya, V. Gorin, and S. Sodin. How weak are weak factors? uniform inference for signal strength in signal plus noise models, 2025. arXiv:2507.18554.
- [20] M. Capitaine. Limiting eigenvectors of outliers for spiked information-plus-noise type matrices. In *Séminaire de Probabilités XLIX*, volume 2215 of *Lecture Notes in Math.*, pages 119–164. Springer, Cham, 2018.
- [21] M. Capitaine and C. Donati-Martin. Non universality of fluctuations of outlier eigenvectors for block diagonal deformations of Wigner matrices. *ALEA Lat. Am. J. Probab. Math. Stat.*, 18(1):129–165, 2021.
- [22] M. Capitaine, C. Donati-Martin, and D. Féral. The largest eigenvalues of finite rank deformation of large Wigner matrices: convergence and nonuniversality of the fluctuations. *Ann. Probab.*, 37(1):1–47, 2009.
- [23] P. B. Denton, S. J. Parke, T. Tao, and X. Zhang. Eigenvectors from eigenvalues: a survey of a basic identity in linear algebra, 2019. arXiv:1908.03795.
- [24] X. Ding. High dimensional deformed rectangular matrices with applications in matrix denoising. *Bernoulli*, 26(1):387–417, 2020.
- [25] I. Dumitriu. *Eigenvalue statistics for beta-ensembles*. ProQuest LLC, Ann Arbor, MI, 2003. Thesis (Ph.D.)–Massachusetts Institute of Technology.
- [26] I. Dumitriu and A. Edelman. Matrix models for beta ensembles. *J. Math. Phys.*, 43(11):5830–5847, 2002.
- [27] D. Féral and S. Péché. The largest eigenvalue of rank one deformation of large Wigner matrices. *Comm. Math. Phys.*, 272(1):185–228, 2007.
- [28] P. J. Forrester and E. M. Rains. Interrelationships between orthogonal, unitary and symplectic matrix ensembles. In *Random matrix models and their applications*, volume 40 of *Math. Sci. Res. Inst. Publ.*, pages 171–207. Cambridge Univ. Press, Cambridge, 2001.
- [29] P. Y. Gaudreau Lamarre, P. Ghosal, W. Li, and Y. Liao. Rigidity of the stochastic Airy operator. *Int. Math. Res. Not. IMRN*, (24):20701–20724, 2023.
- [30] D. A. Harville. *Matrix algebra from a statistician's perspective*. Springer-Verlag, New York, 1997.
- [31] A. Knowles and J. Yin. The isotropic semicircle law and deformation of Wigner matrices. *Comm. Pure Appl. Math.*, 66(11):1663–1750, 2013.
- [32] A. Knowles and J. Yin. The outliers of a deformed Wigner matrix. *Ann. Probab.*, 42(5):1980–2031, 2014.
- [33] B. Landon and T. Stojimirovic. Multilevel limits of spiked random matrix minors, 2026. arXiv:2608.01531.
- [34] E. S. Meckes. *The random matrix theory of the classical compact groups*, volume 218 of *Cambridge Tracts in Mathematics*. Cambridge University Press, Cambridge, 2019.
- [35] M. Y. Mo. Rank 1 real Wishart spiked model. *Comm. Pure Appl. Math.*, 65(11):1528–1638, 2012.
- [36] S. O’Rourke. Gaussian fluctuations of eigenvalues in Wigner random matrices. *J. Stat. Phys.*, 138(6):1045–1066, 2010.
- [37] D. Paul. Asymptotics of sample eigenstructure for a large dimensional spiked covariance model. *Statist. Sinica*, 17(4):1617–1642, 2007.
- [38] S. Péché. The largest eigenvalue of small rank perturbations of Hermitian random matrices. *Probab. Theory Related Fields*, 134(1):127–173, 2006.
- [39] J. A. Ramírez, B. Rider, and B. Virág. Beta ensembles, stochastic Airy spectrum, and a diffusion. *J. Amer. Math. Soc.*, 24(4):919–944, 2011.
- [40] A. B. Soshnikov. Gaussian fluctuation for the number of particles in Airy, Bessel, sine, and other determinantal random point fields. *J. Statist. Phys.*, 100(3-4):491–522, 2000.
- [41] T. Tao and V. Vu. Random matrices have simple spectrum. *Combinatorica*, 37(3):539–553, 2017.
- [42] C. Zhong. Large deviation bounds for the Airy point process, 2019. arXiv:1910.00797.